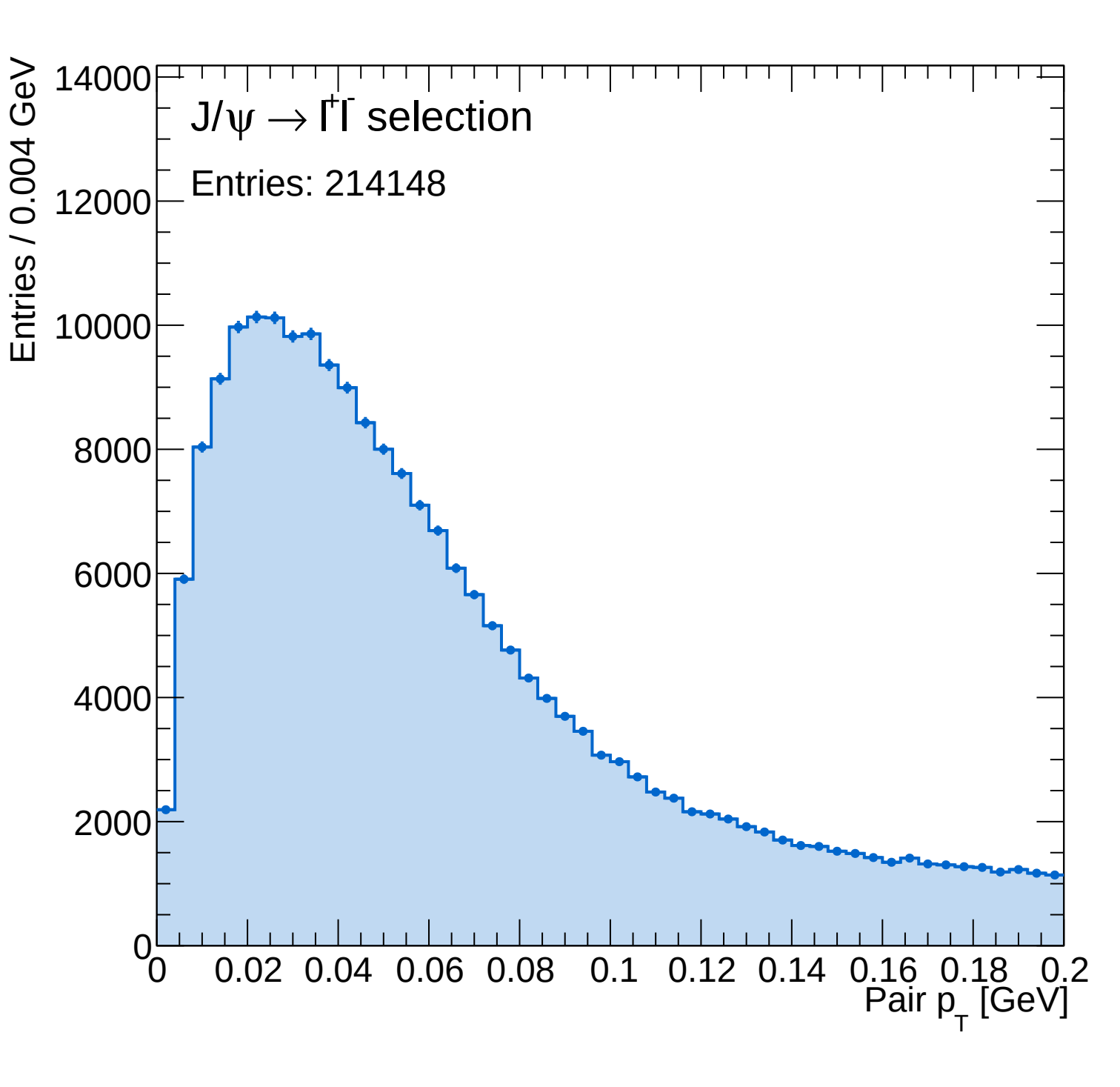
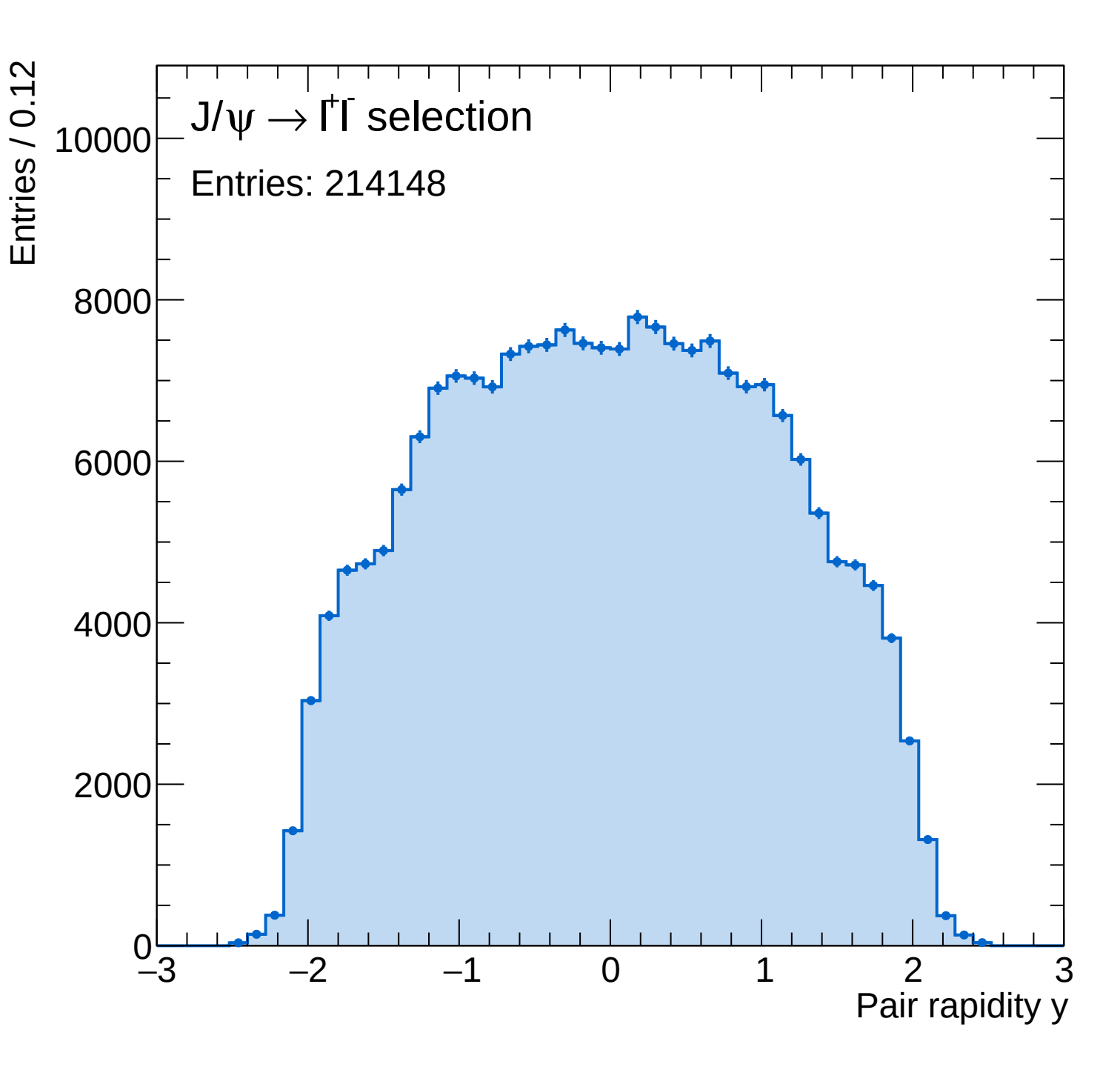
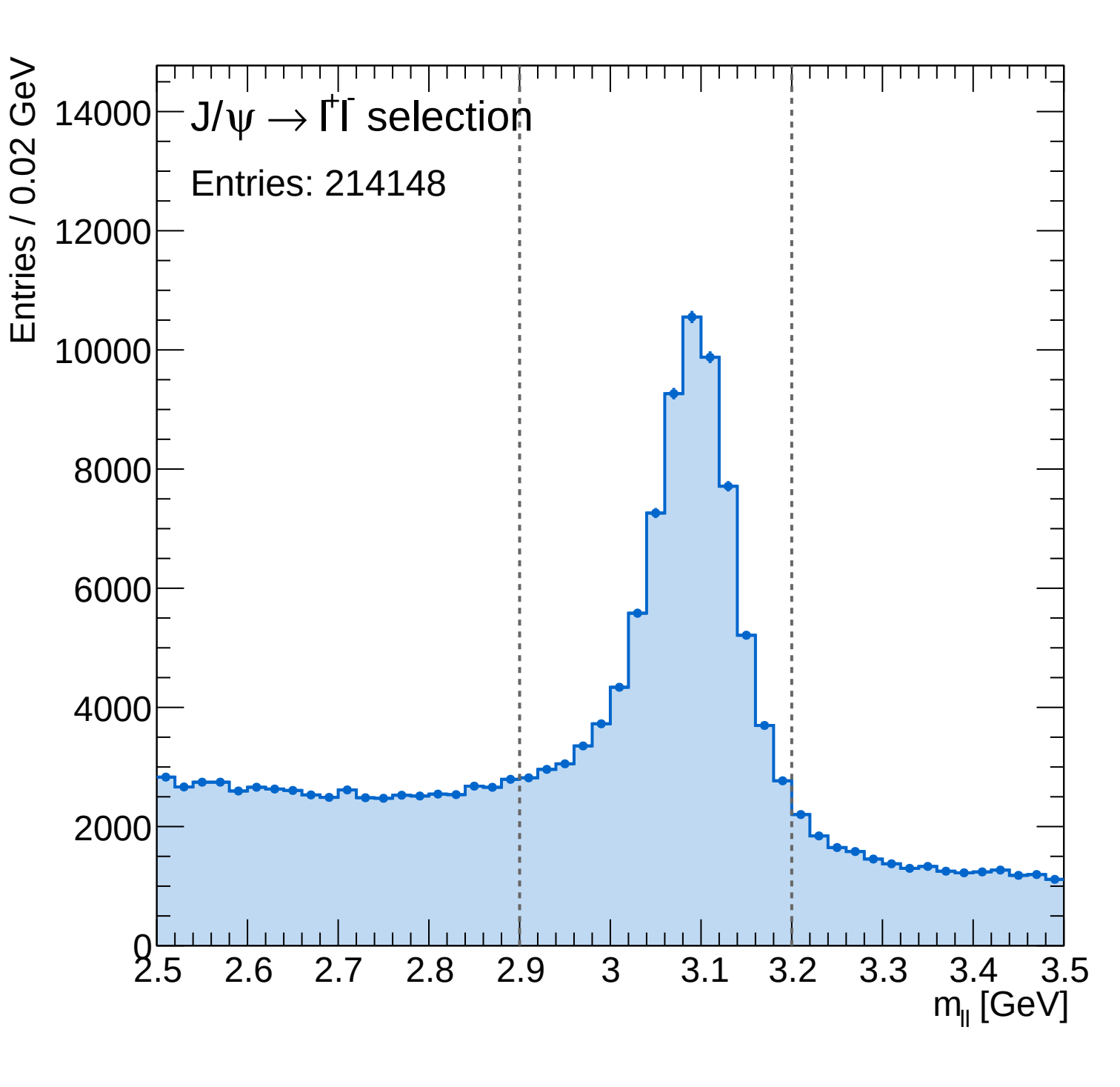
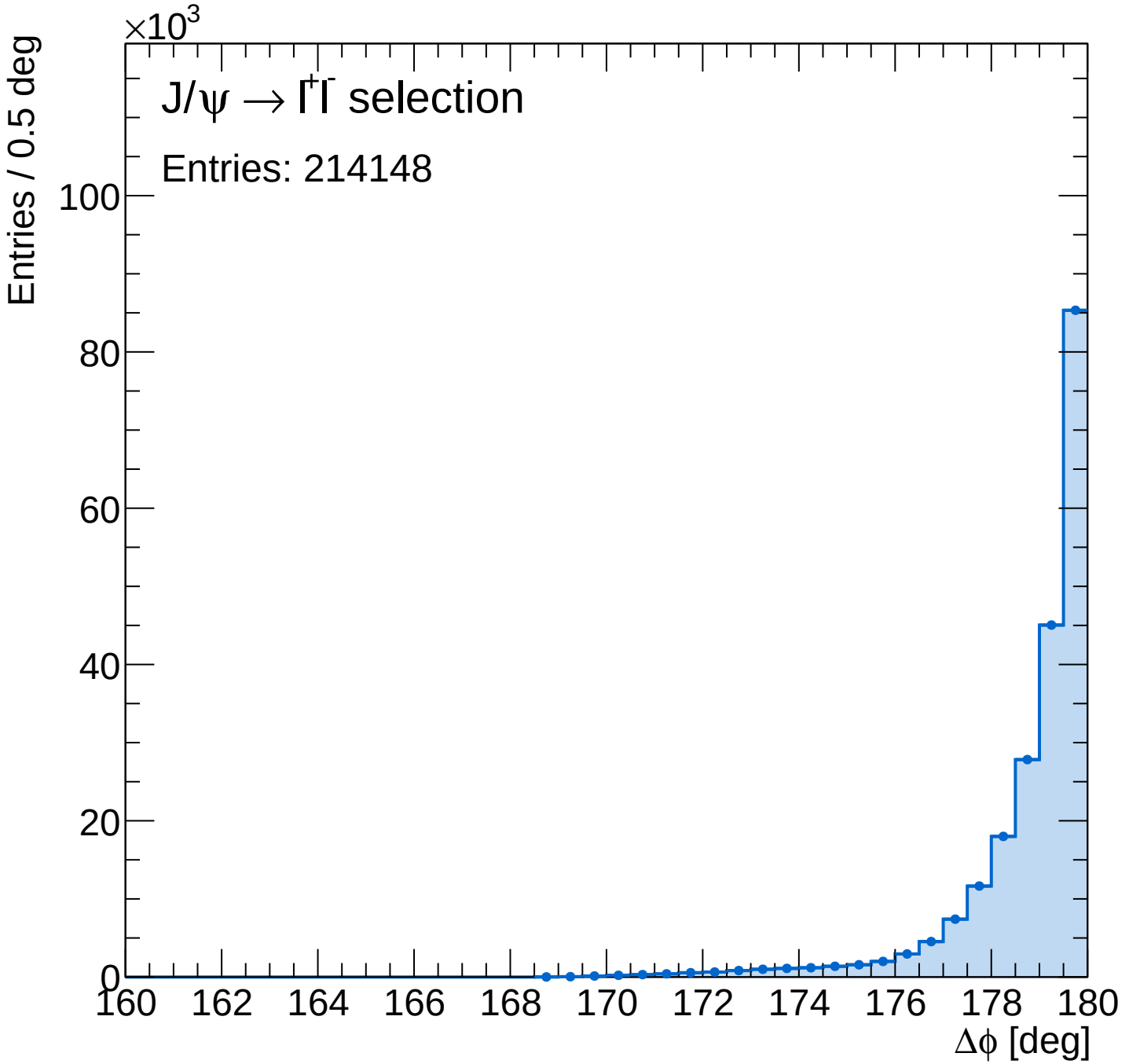
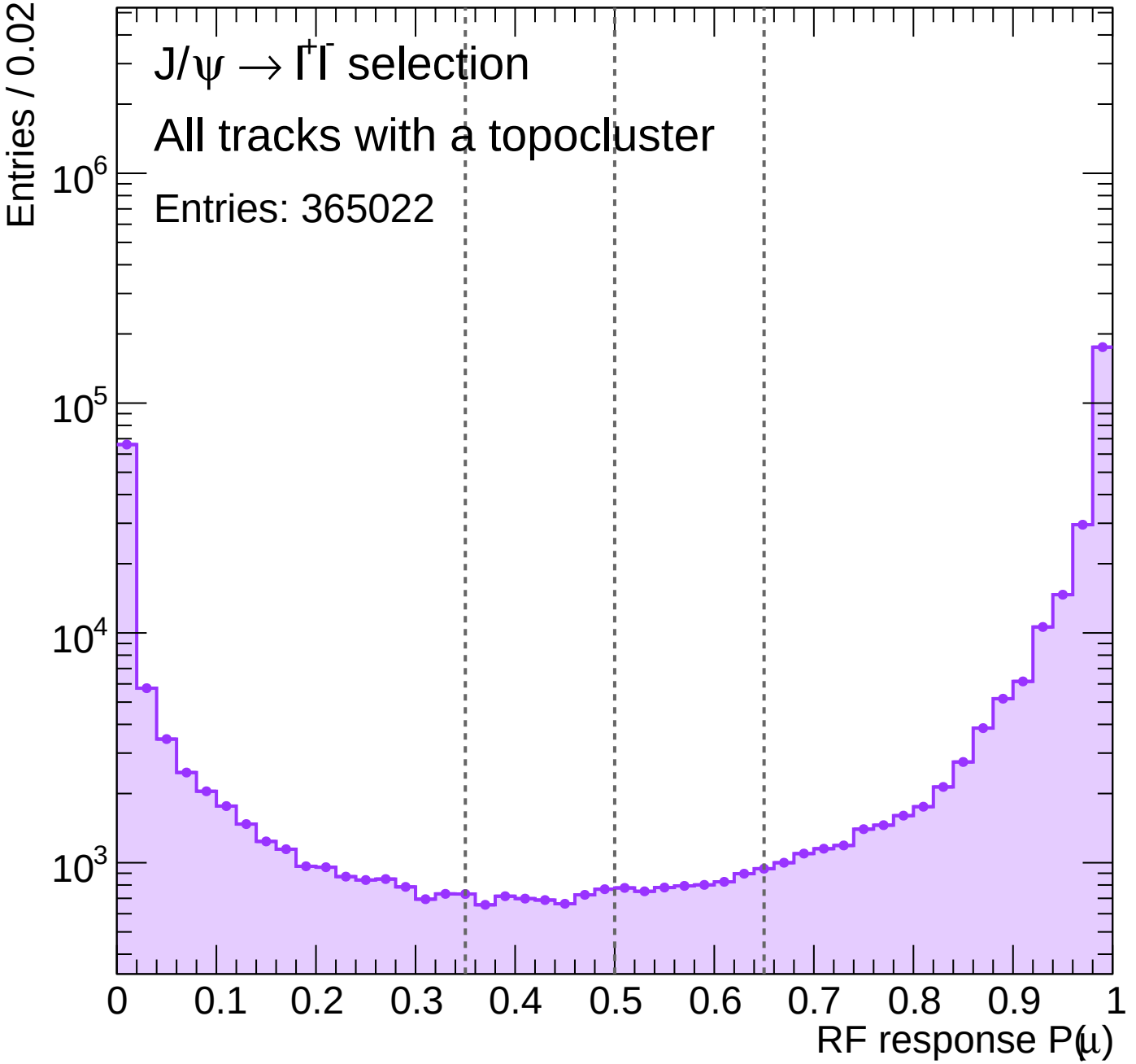
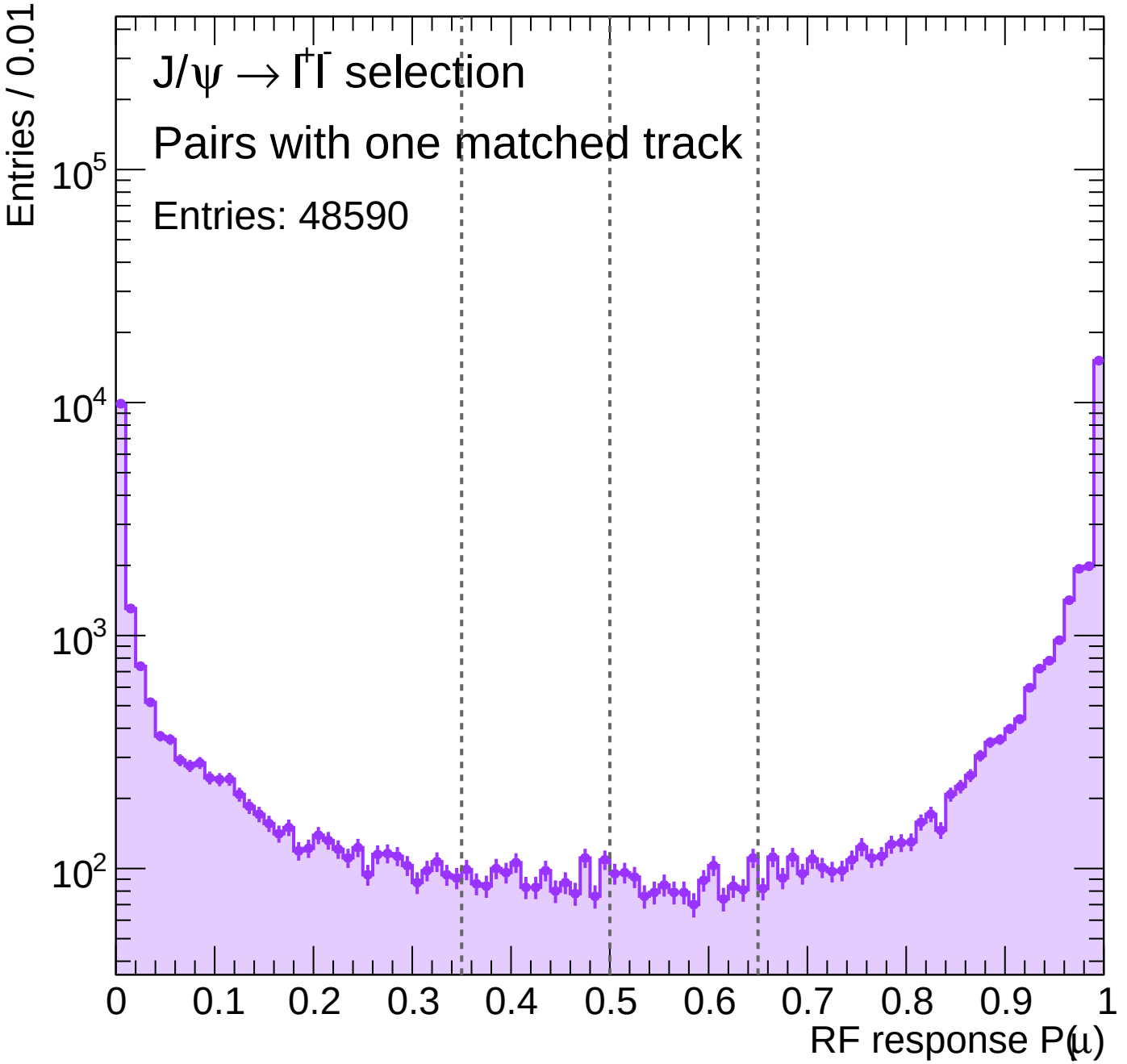

J/ψ → l⁺l⁻ selection
Entries: 214148
Entries / 0.004 GeV
14000
12000
10000
8000
6000
4000
2000
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
Pair p_T [GeV]

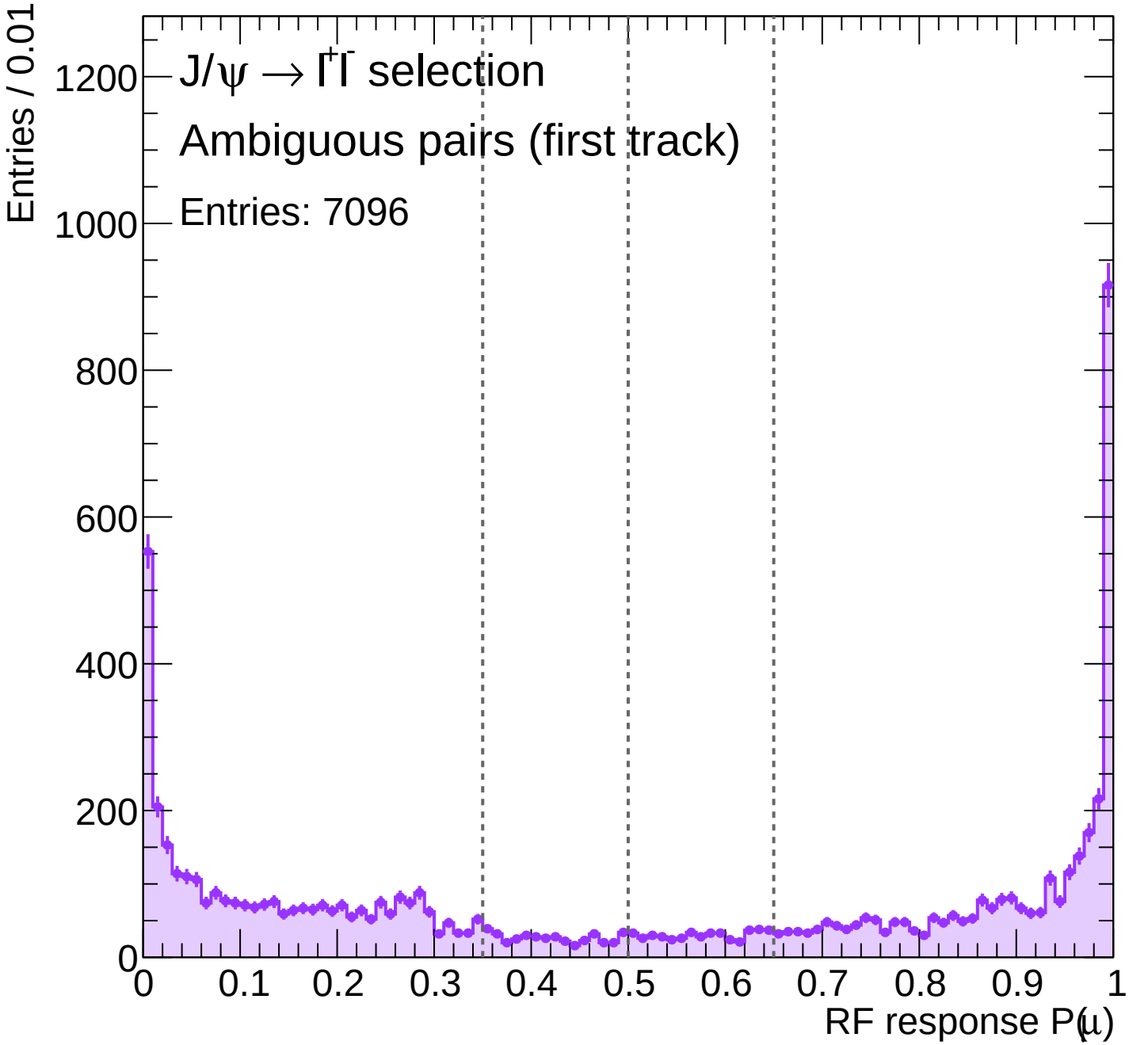

J/ψ → l⁺l⁻ selection
Entries: 214148
Entries / 0.12
10000
8000
6000
4000
2000
0
–3
–2
–1
0
1
2
3
Pair rapidity y

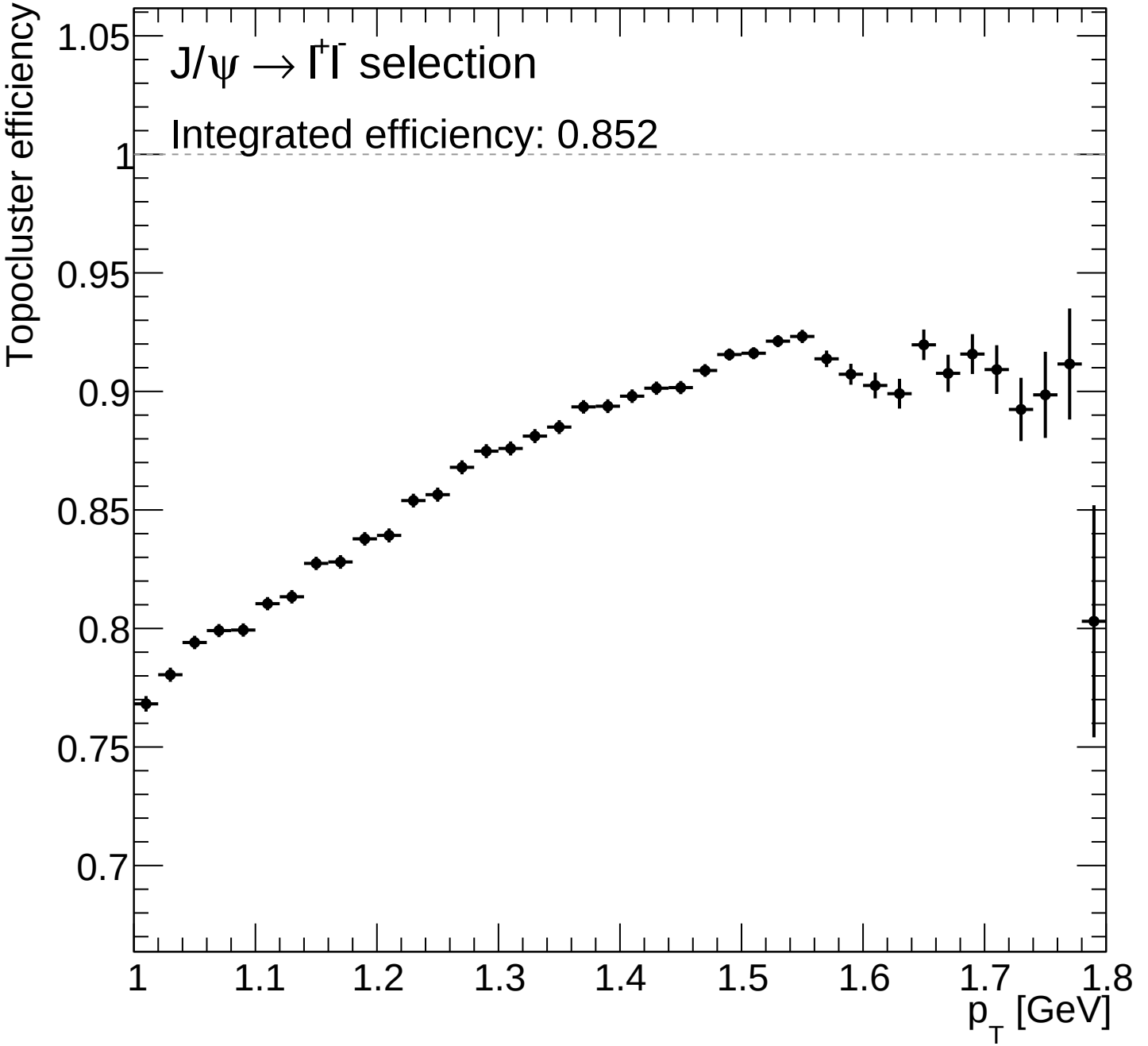

J/ψ → l⁺l⁻ selection
Entries: 214148
Entries / 0.02 GeV
14000
12000
10000
8000
6000
4000
2000
0
2.5
2.6
2.7
2.8
2.9
3
3.1
3.2
3.3
3.4
3.5
m_ll [GeV]

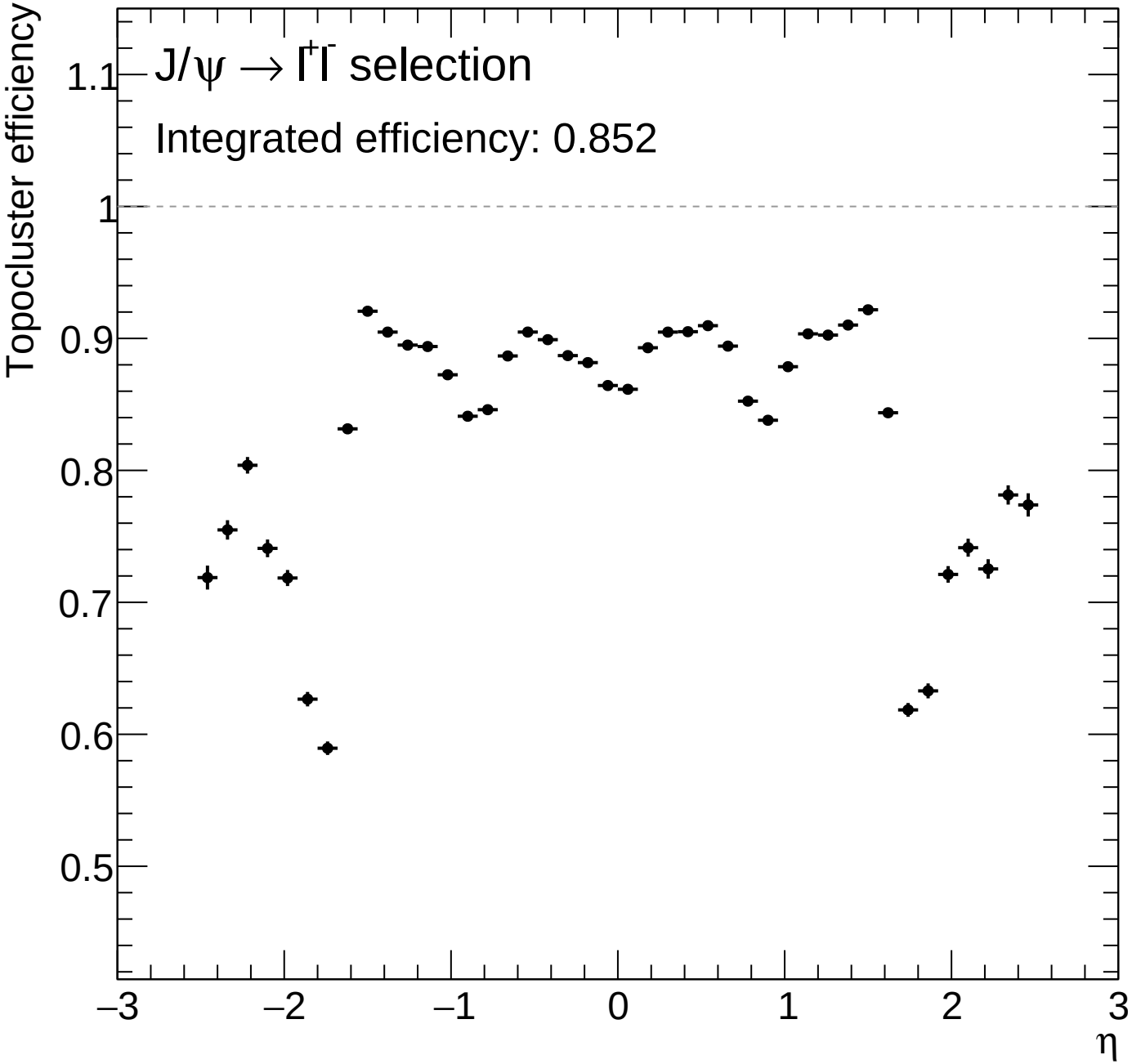

$\times 10^3$
$J/\psi \to l^+l^-$ selection
Entries: 214148
Entries / 0.5 deg
100
80
60
40
20
0
160
162
164
166
168
170
172
174
176
178
180
$\Delta\phi$ [deg]

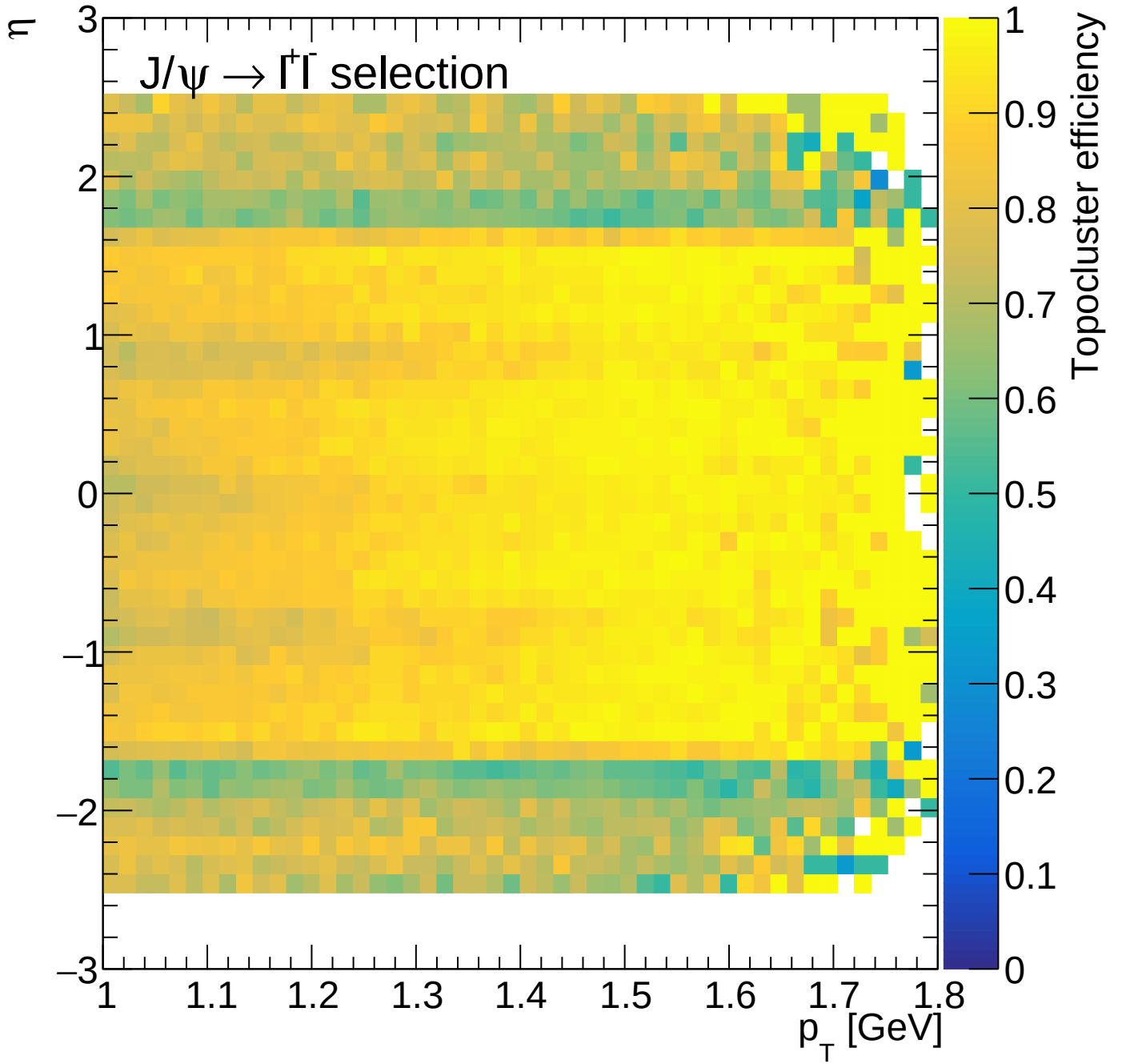

$J/\psi \to l^+l^-$ selection
All tracks with a topocluster
Entries: 365022
Entries / 0.02
10^6
10^5
10^4
10^3
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
RF response P(μ)

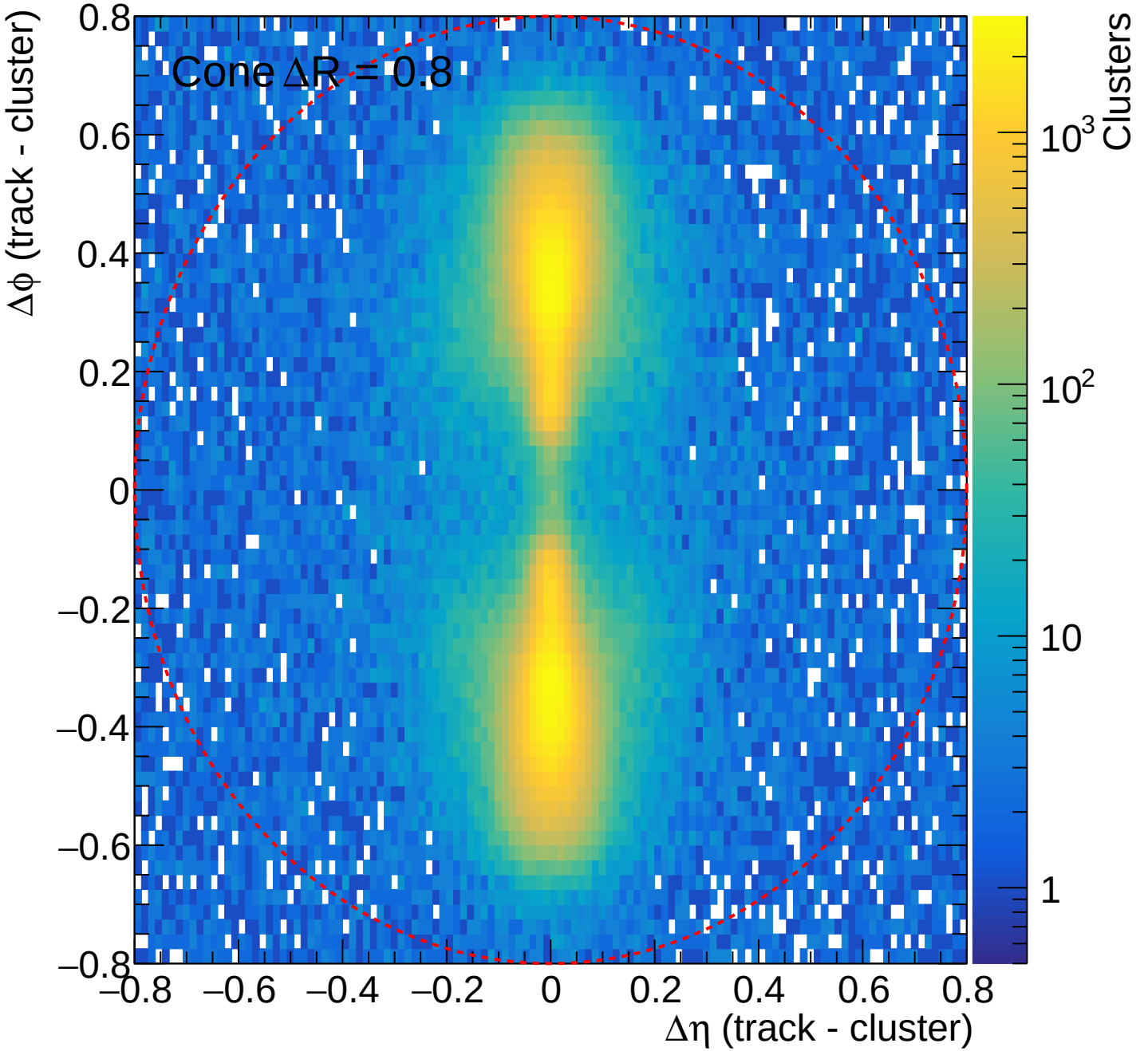

$J/\psi \to l^+l^-$ selection
Pairs with one matched track
Entries: 48590
Entries / 0.01
10^5
10^4
10^3
10^2
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
RF response P(μ)

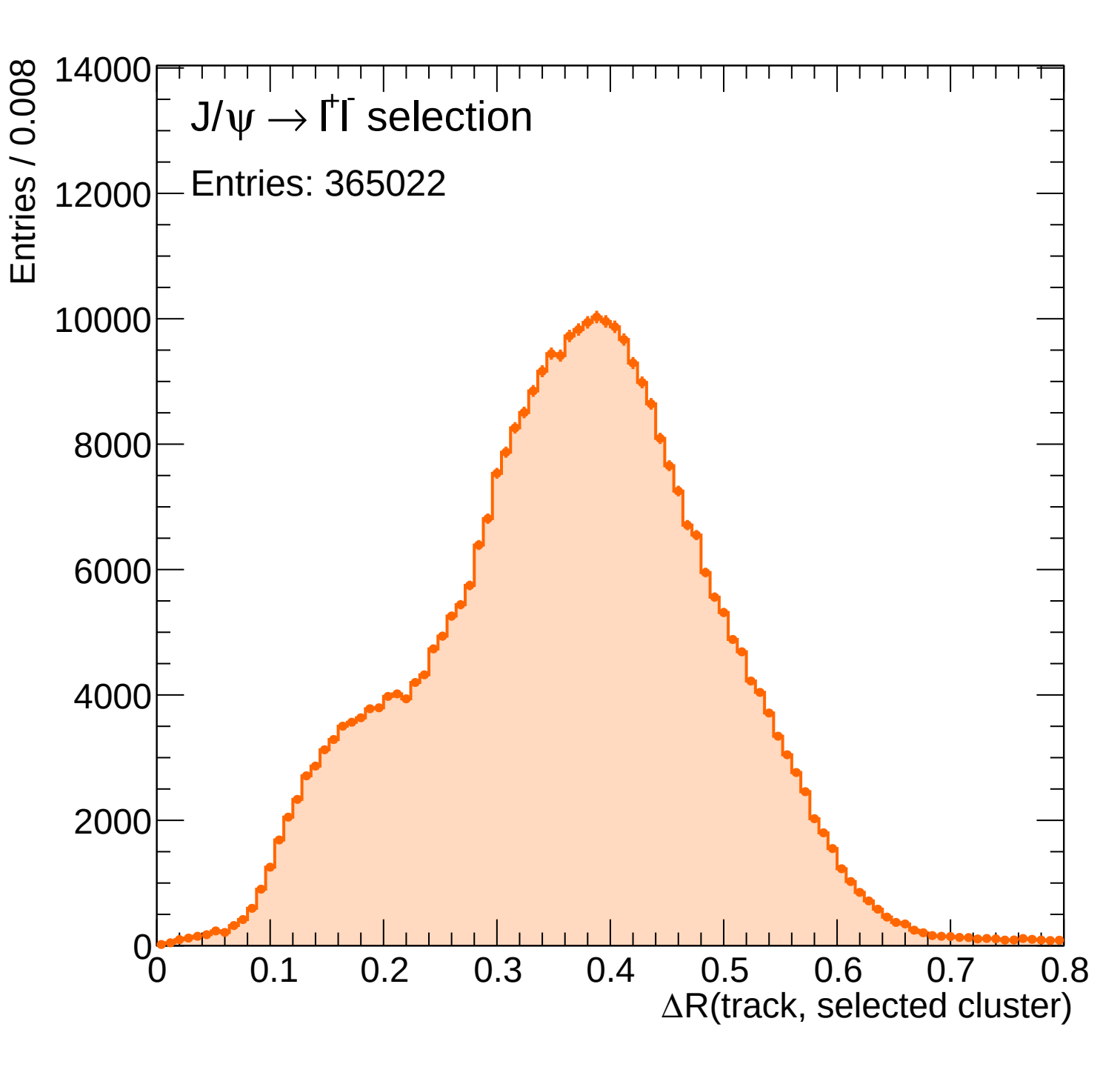

J/ψ → l+l- selection
Ambiguous pairs (first track)
Entries: 7096
Entries / 0.01
0
200
400
600
800
1000
1200
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
RF response P(μ)

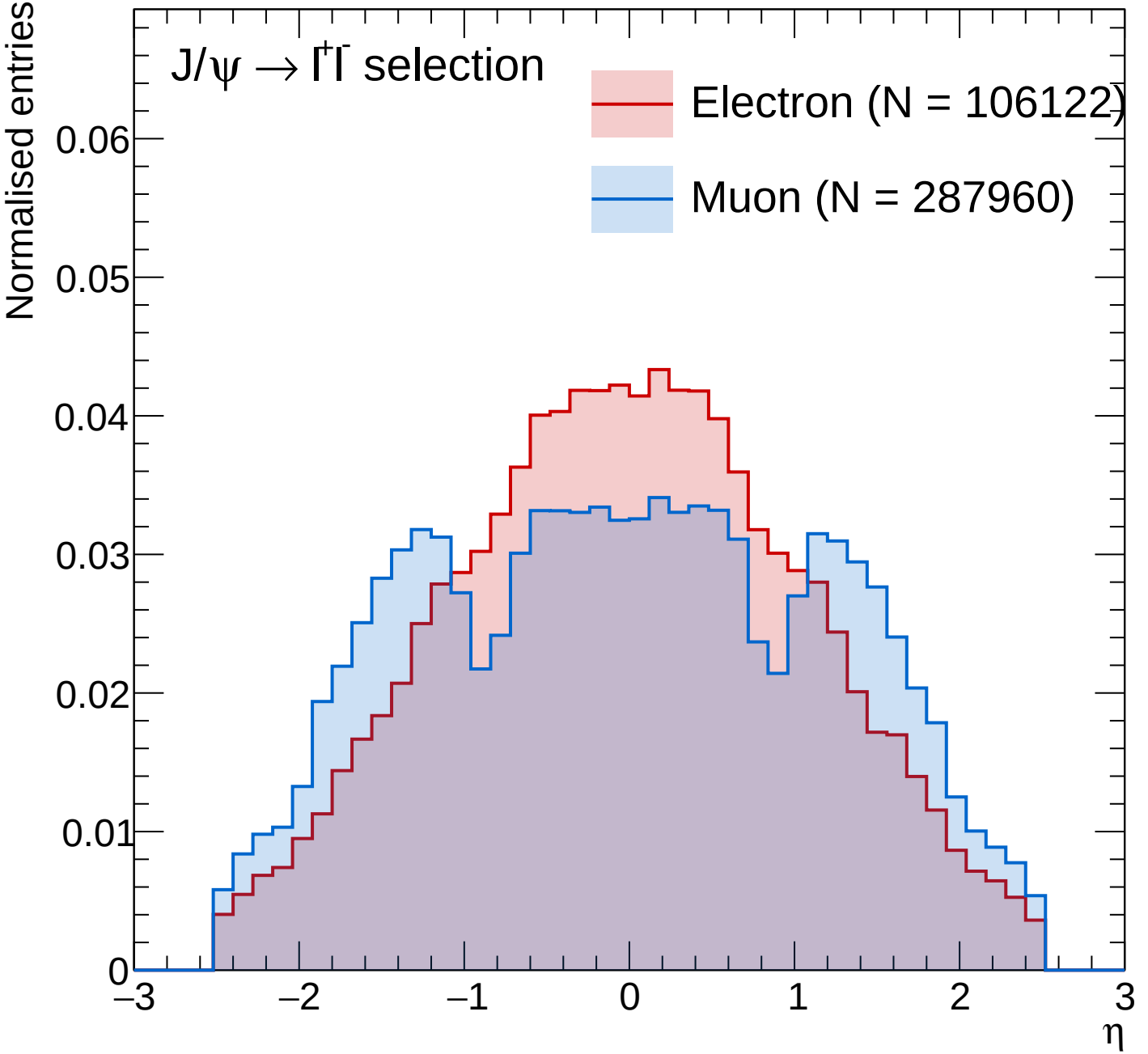

$J/\psi \to l^+l^-$ selection
Integrated efficiency: 0.852
Topocluster efficiency
1.05
1
0.95
0.9
0.85
0.8
0.75
0.7
1
1.1
1.2
1.3
1.4
1.5
1.6
1.7
1.8
p_T [GeV]

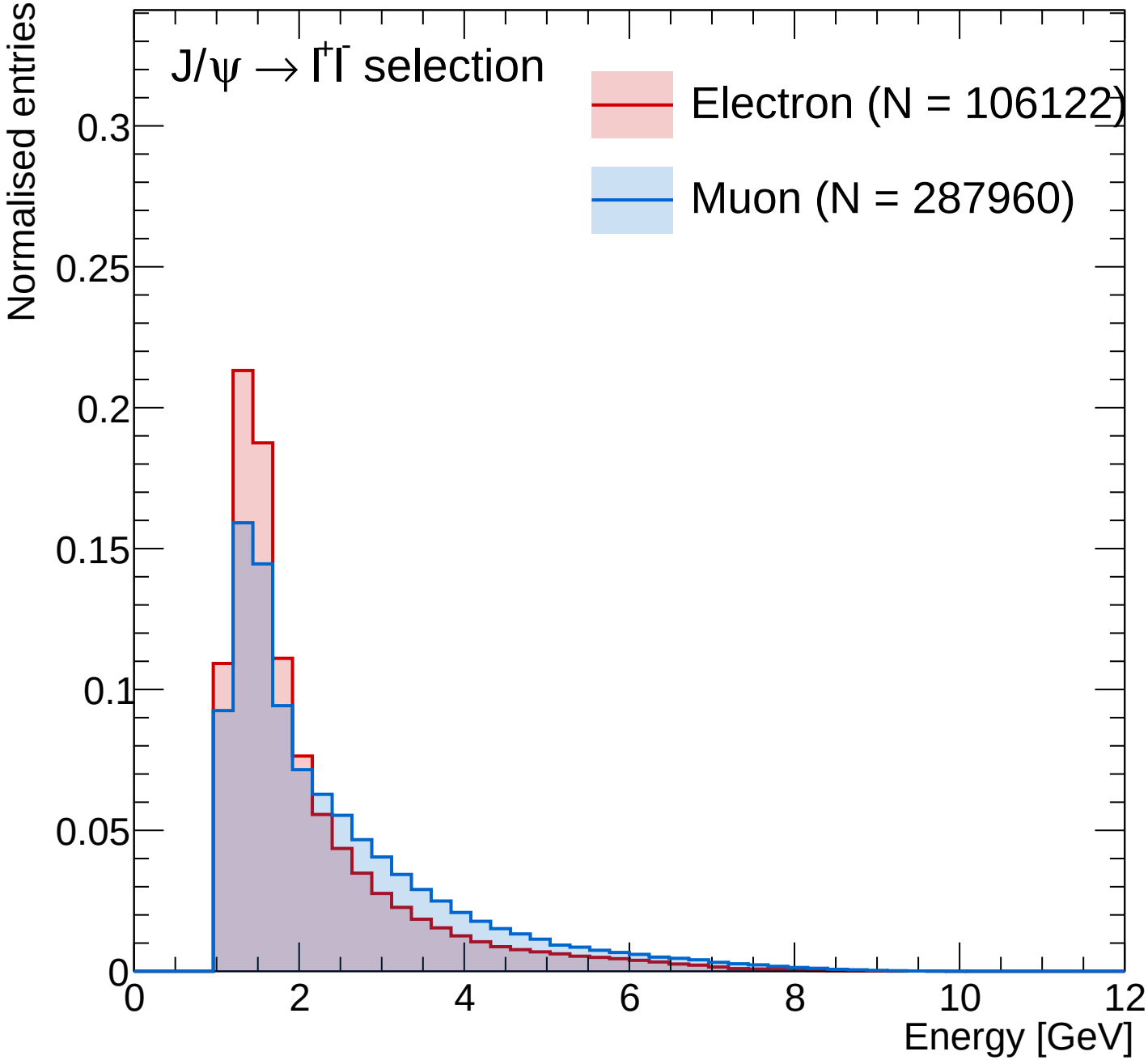

$J/\psi \to l^+l^-$ selection
Integrated efficiency: 0.852
Topocluster efficiency
1.1
1
0.9
0.8
0.7
0.6
0.5
–3
–2
–1
0
1
2
3
η

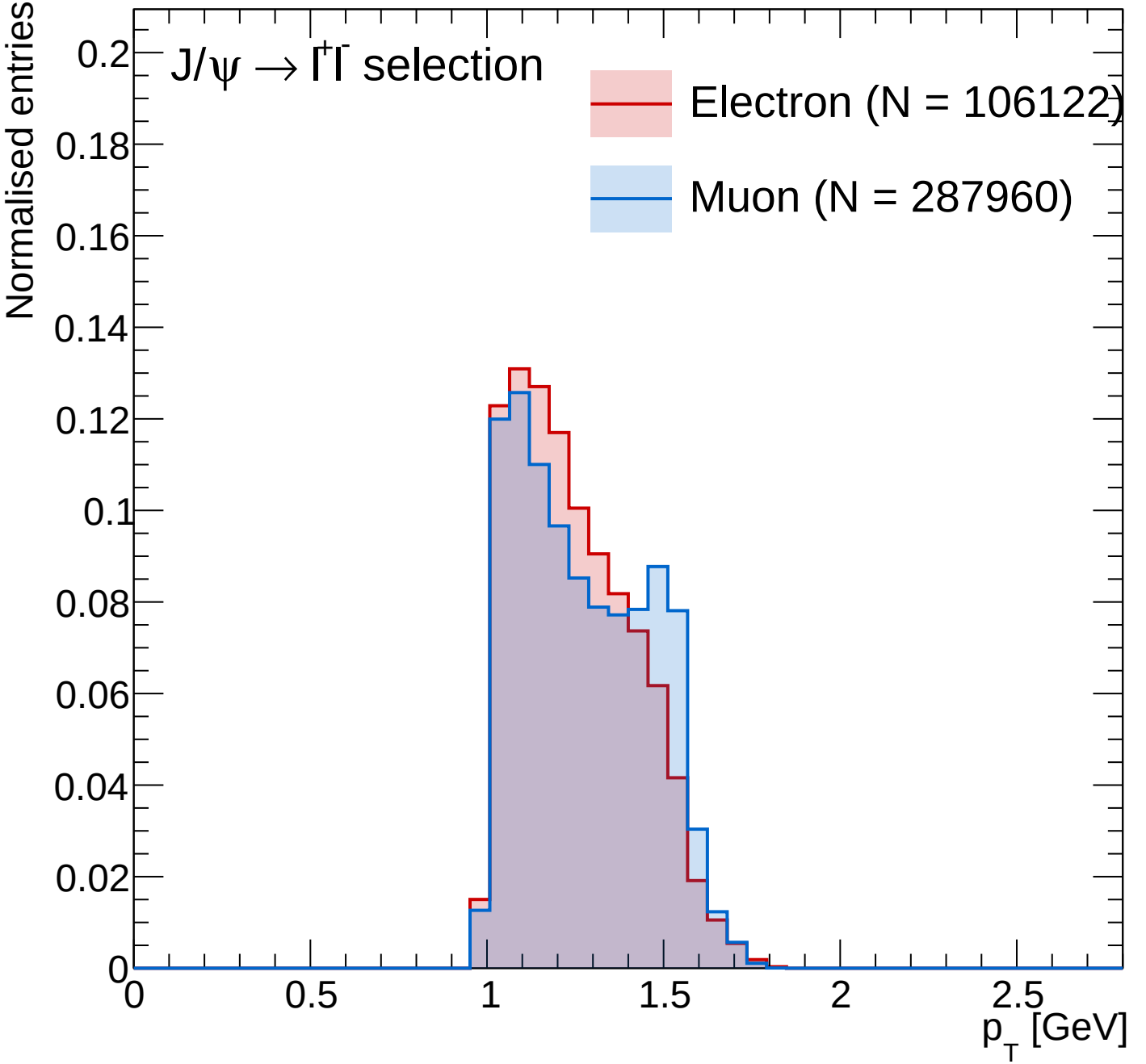

J/ψ → l⁺l⁻ selection
η
p_T [GeV]
Topocluster efficiency
3
2
1
0
−1
−2
−3
1
1.1
1.2
1.3
1.4
1.5
1.6
1.7
1.8
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0

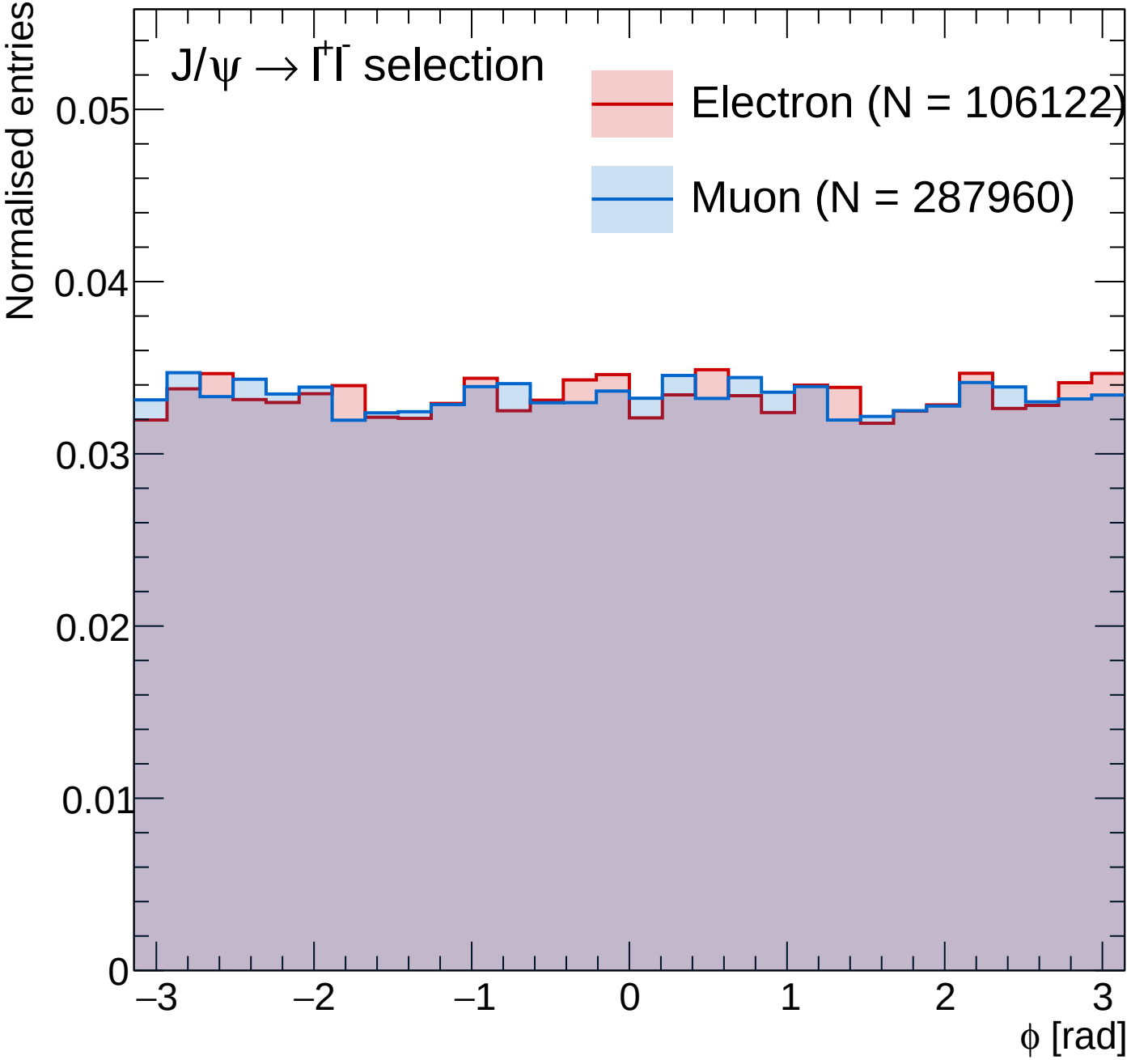

Cone ΔR = 0.8
Δφ (track - cluster)
Δη (track - cluster)
Clusters
0.8
0.6
0.4
0.2
0
–0.2
–0.4
–0.6
–0.8
–0.8 –0.6 –0.4 –0.2 0 0.2 0.4 0.6 0.8
10^3
10^2
10
1

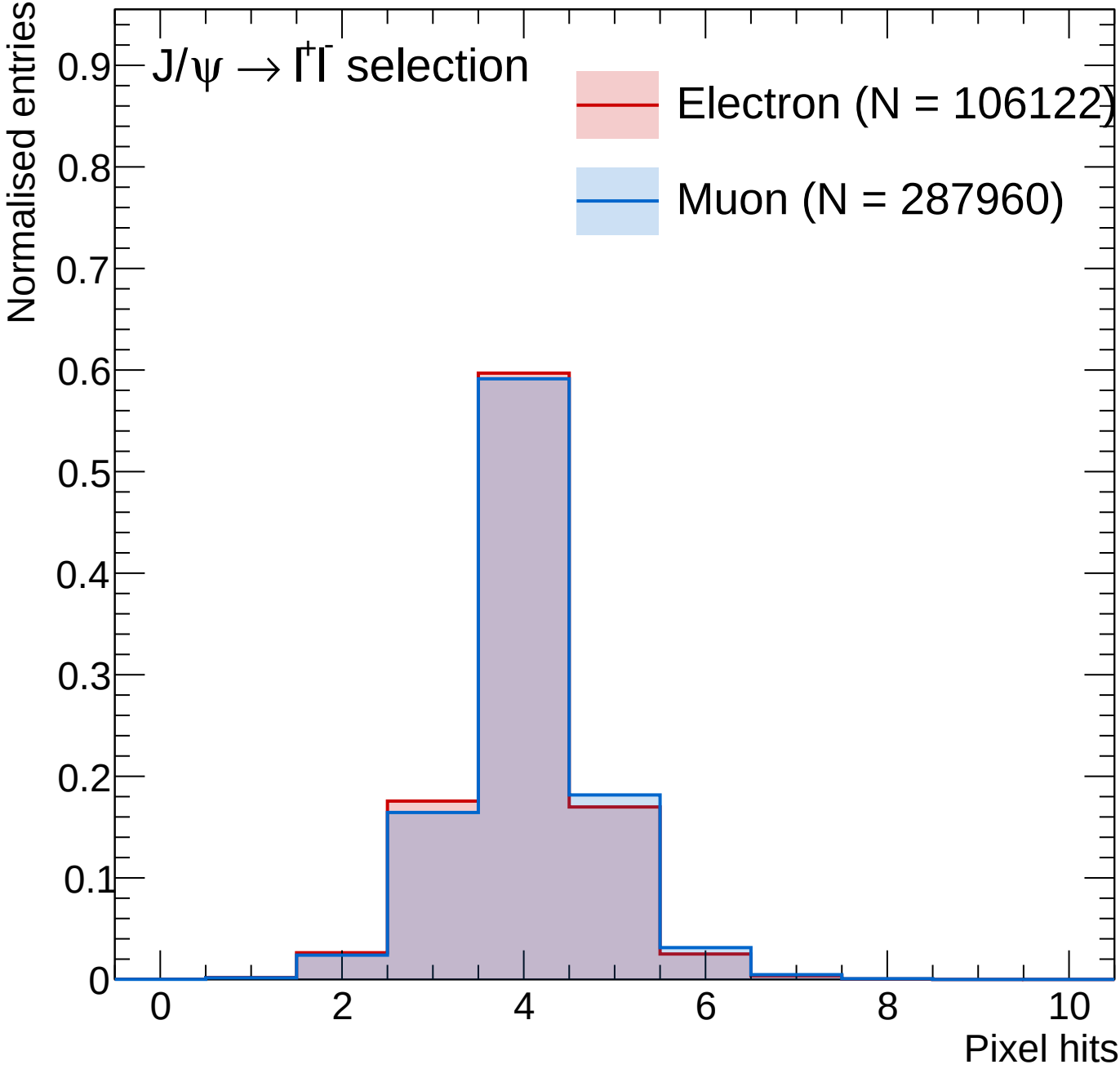

J/ψ → l+l- selection
Entries: 365022
Entries / 0.008
14000
12000
10000
8000
6000
4000
2000
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
ΔR(track, selected cluster)

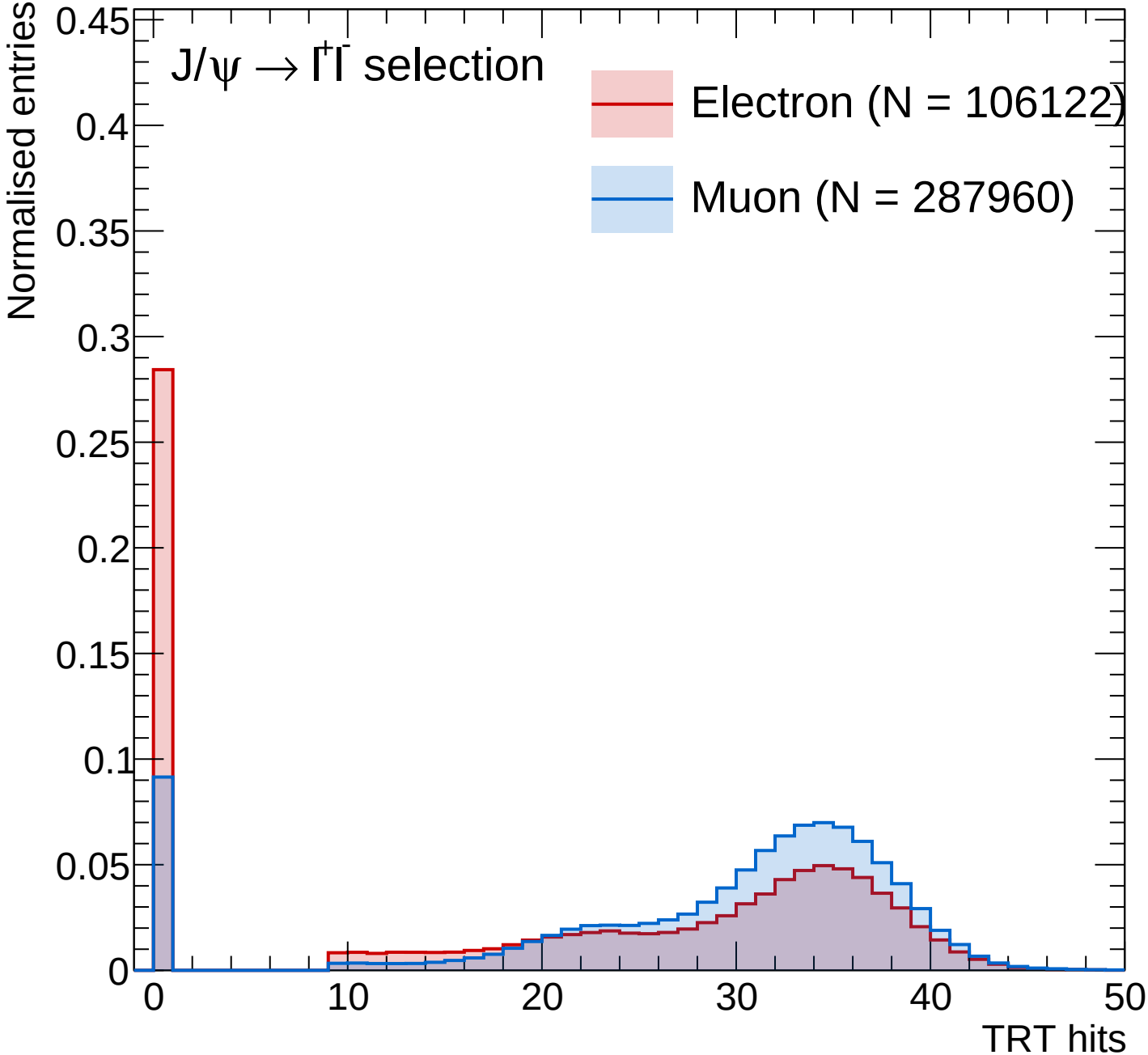

J/ψ → l⁺l⁻ selection
Electron (N = 106122)
Muon (N = 287960)
Normalised entries
0
0.01
0.02
0.03
0.04
0.05
0.06
−3
−2
−1
0
1
2
3
η

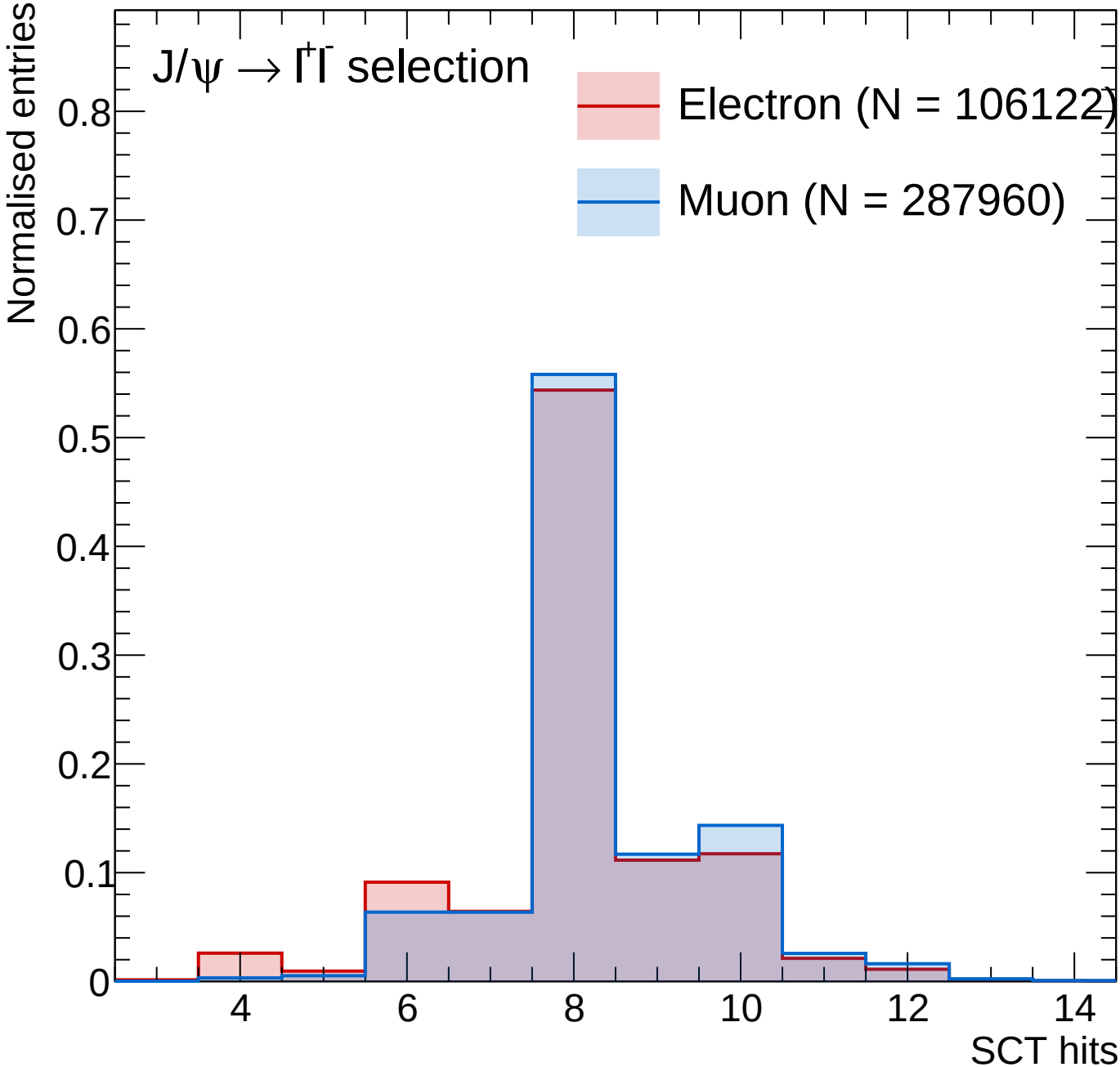

J/ψ → l⁺l⁻ selection
Electron (N = 106122)
Muon (N = 287960)
Normalised entries
0
0.05
0.1
0.15
0.2
0.25
0.3
0
2
4
6
8
10
12
Energy [GeV]

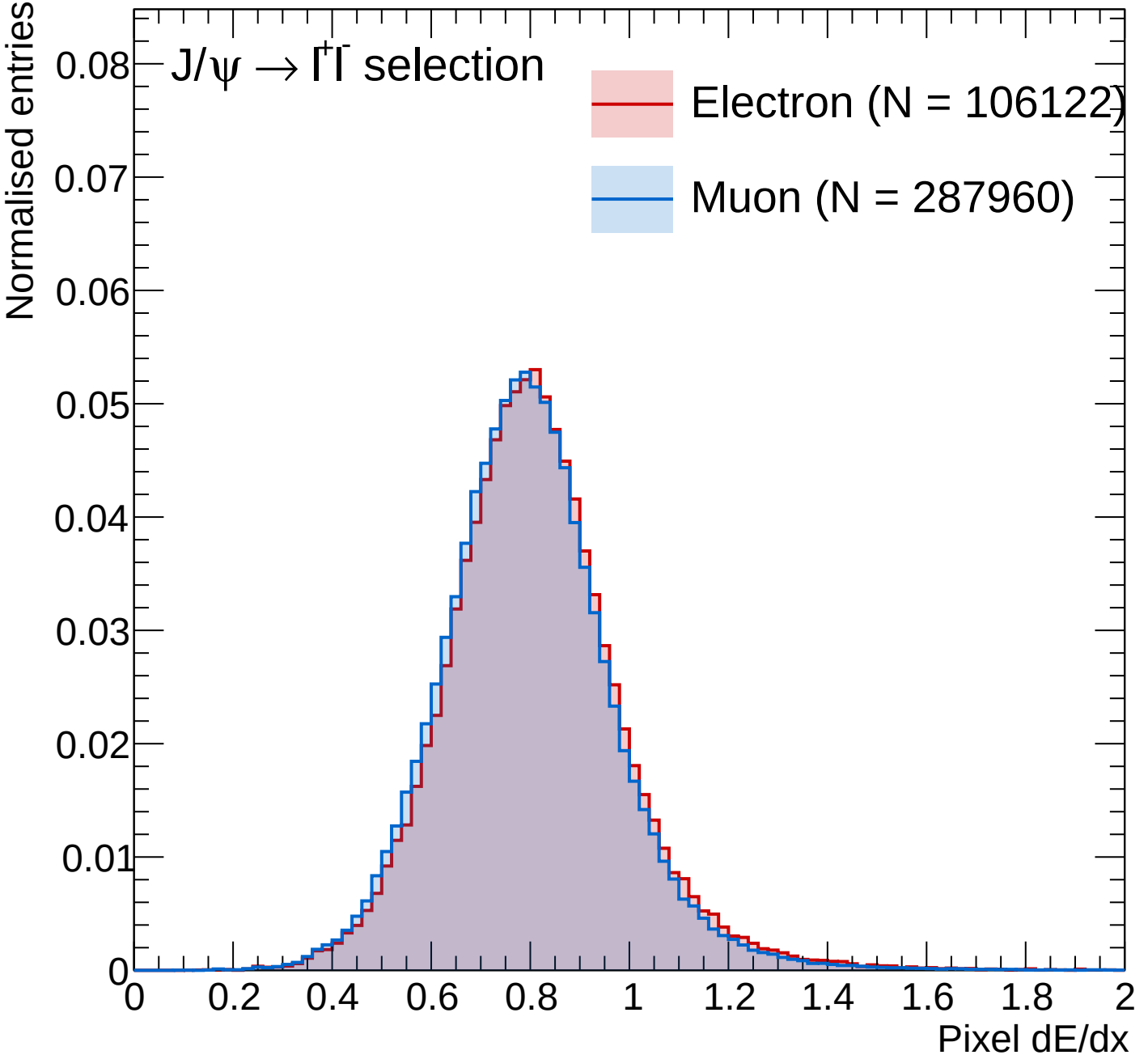

$J/\psi \rightarrow l^+l^-$ selection
Electron (N = 106122)
Muon (N = 287960)
Normalised entries
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.5
1
1.5
2
2.5
p_T [GeV]

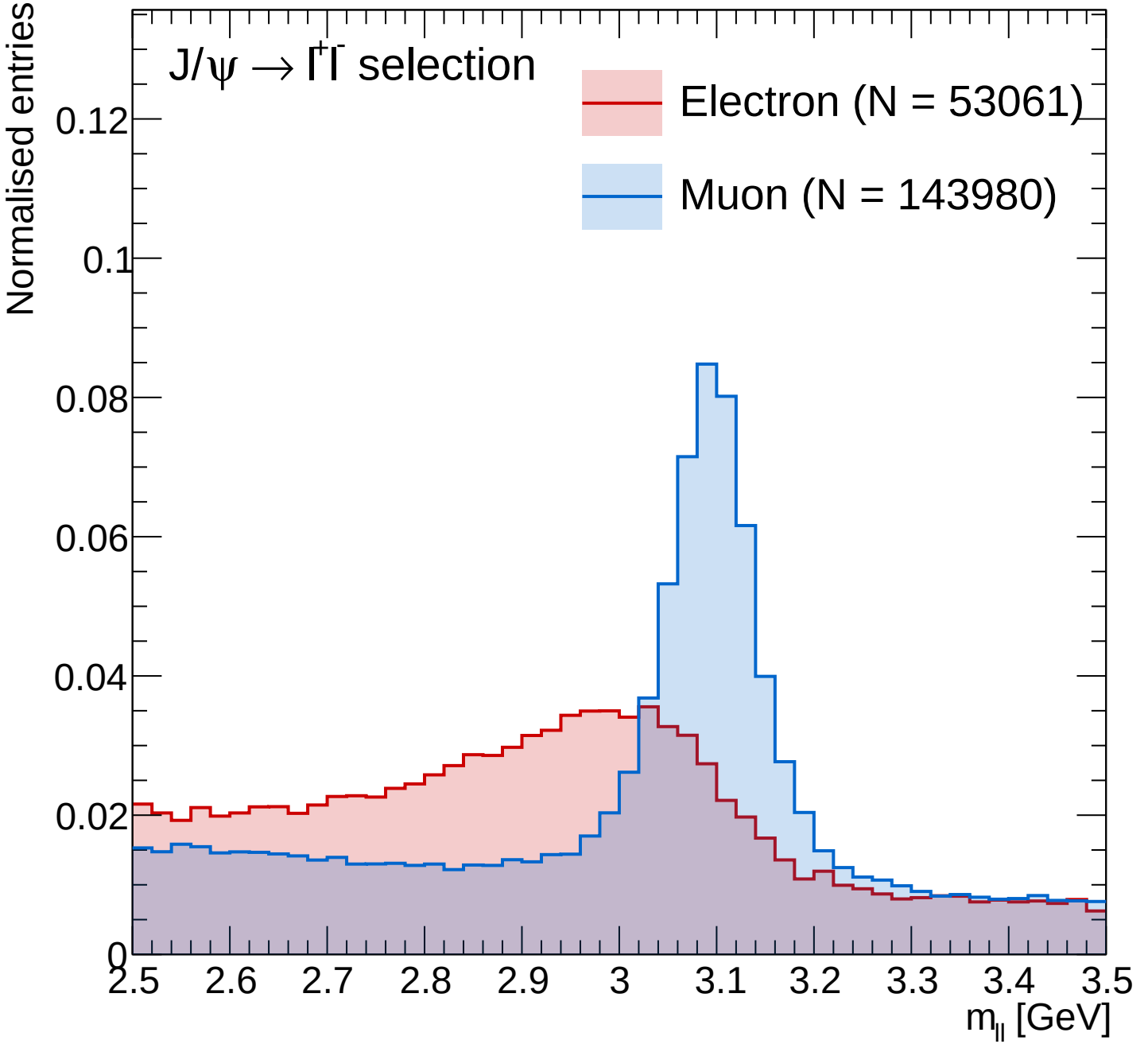

$J/\psi \rightarrow l^+l^-$ selection
Electron (N = 106122)
Muon (N = 287960)
Normalised entries
0.05
0.04
0.03
0.02
0.01
0
–3
–2
–1
0
1
2
3
ϕ [rad]

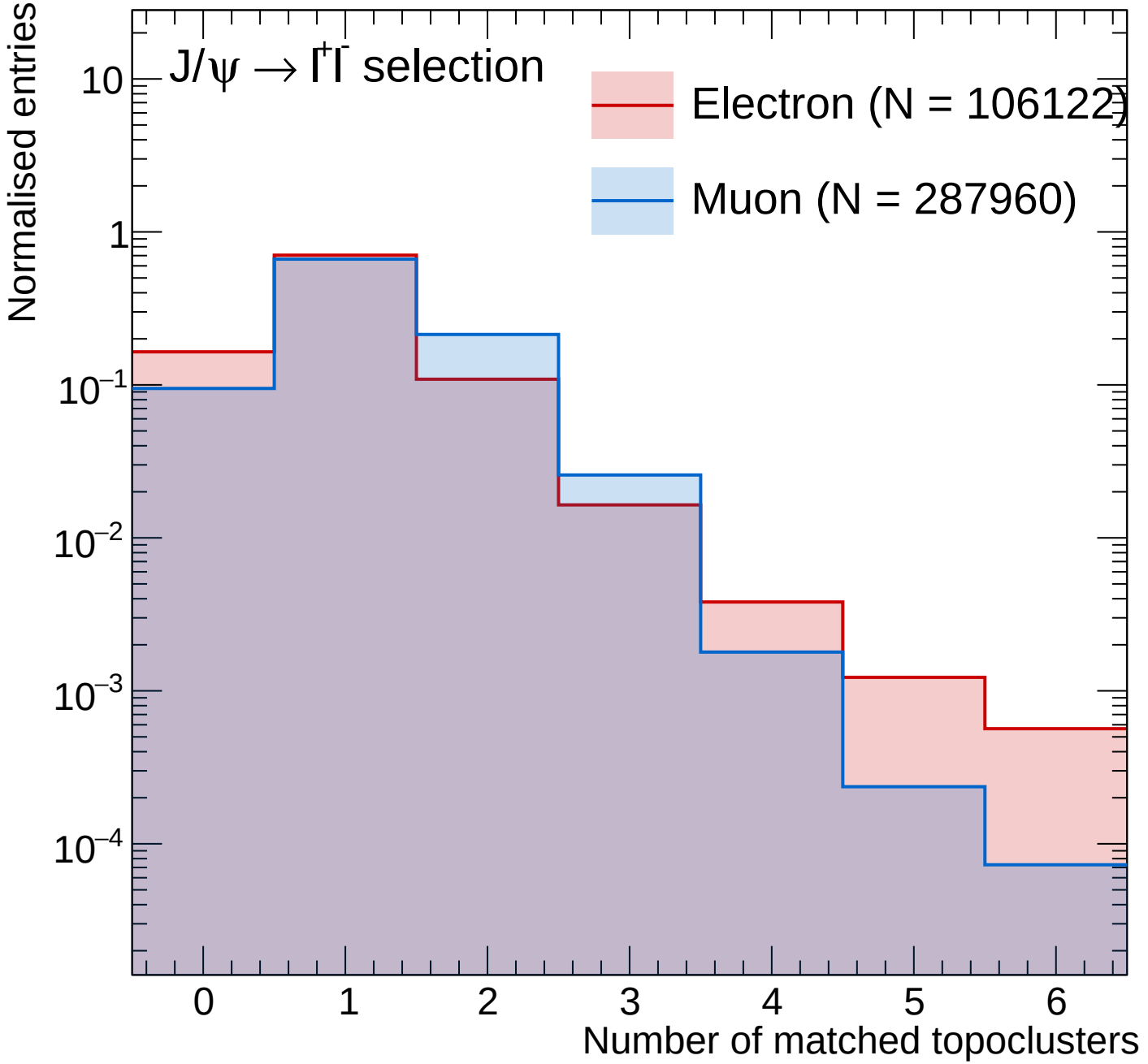

$J/\psi \rightarrow l^+l^-$ selection
Electron (N = 106122)
Muon (N = 287960)
Normalised entries
Pixel hits
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
0
2
4
6
8
10

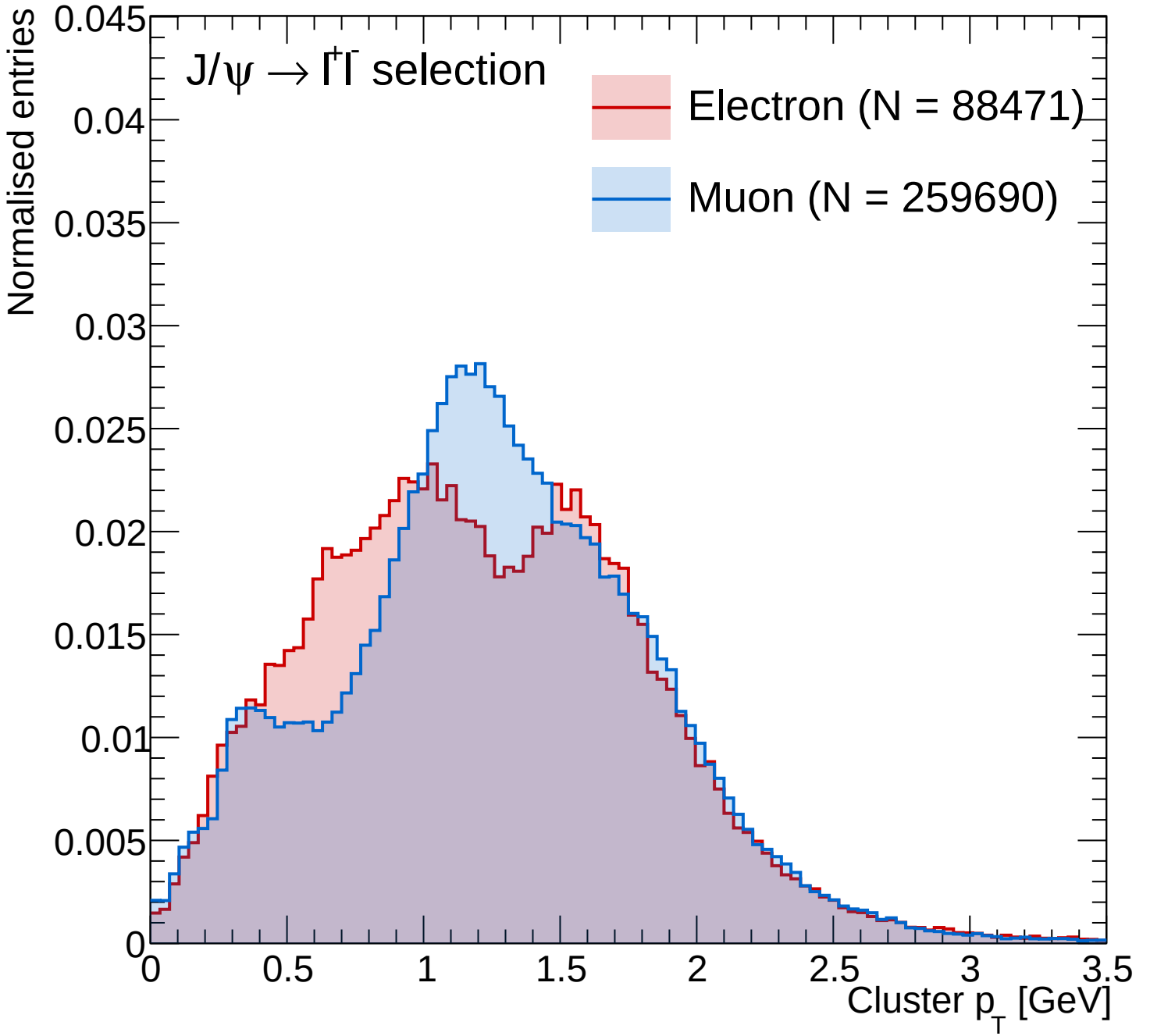

$J/\psi \rightarrow l^+l^-$ selection
Electron (N = 106122)
Muon (N = 287960)
Normalised entries
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
10
20
30
40
50
TRT hits

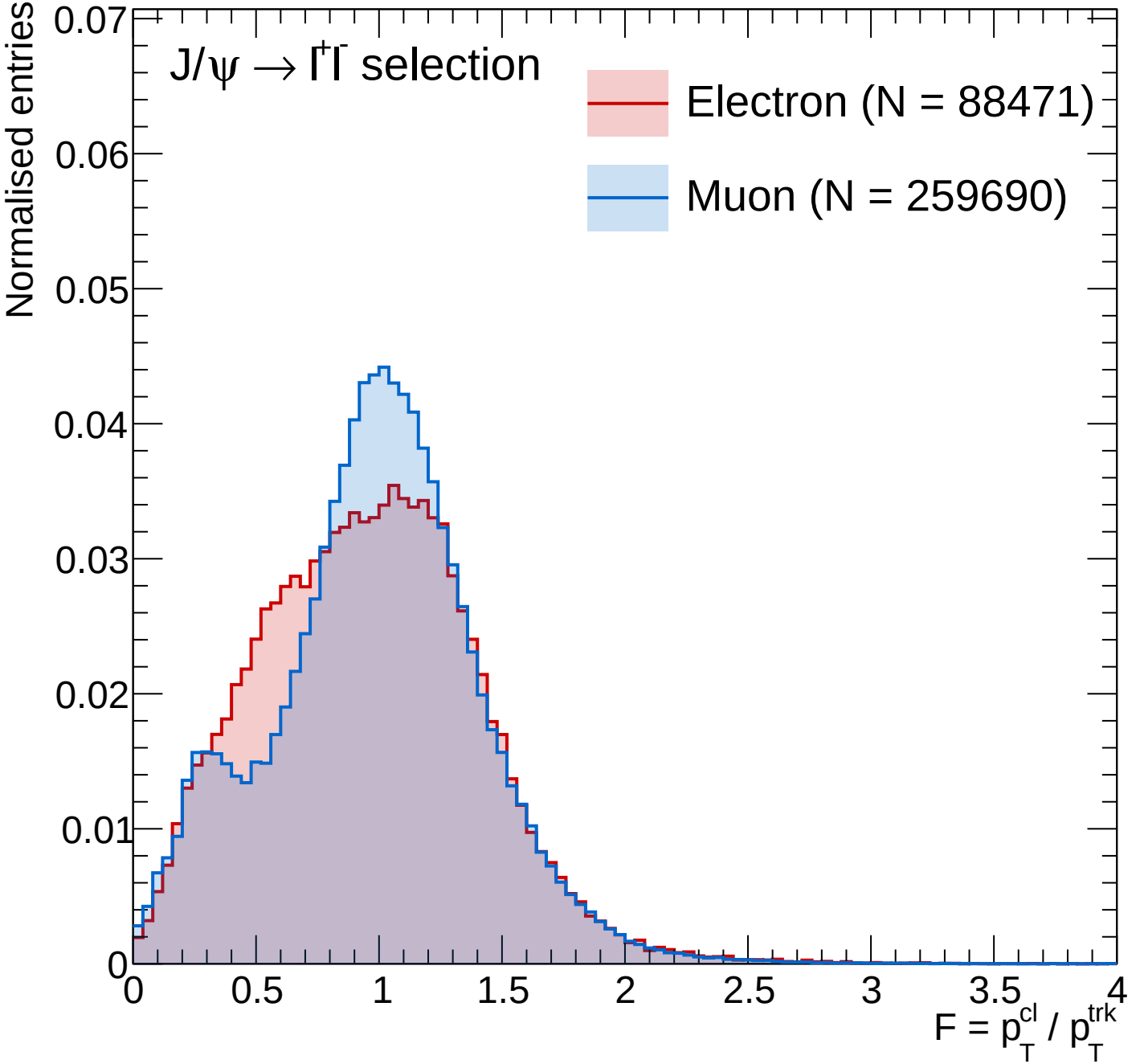

J/ψ → l⁺l⁻ selection
Electron (N = 106122)
Muon (N = 287960)
Normalised entries
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
4
6
8
10
12
14
SCT hits

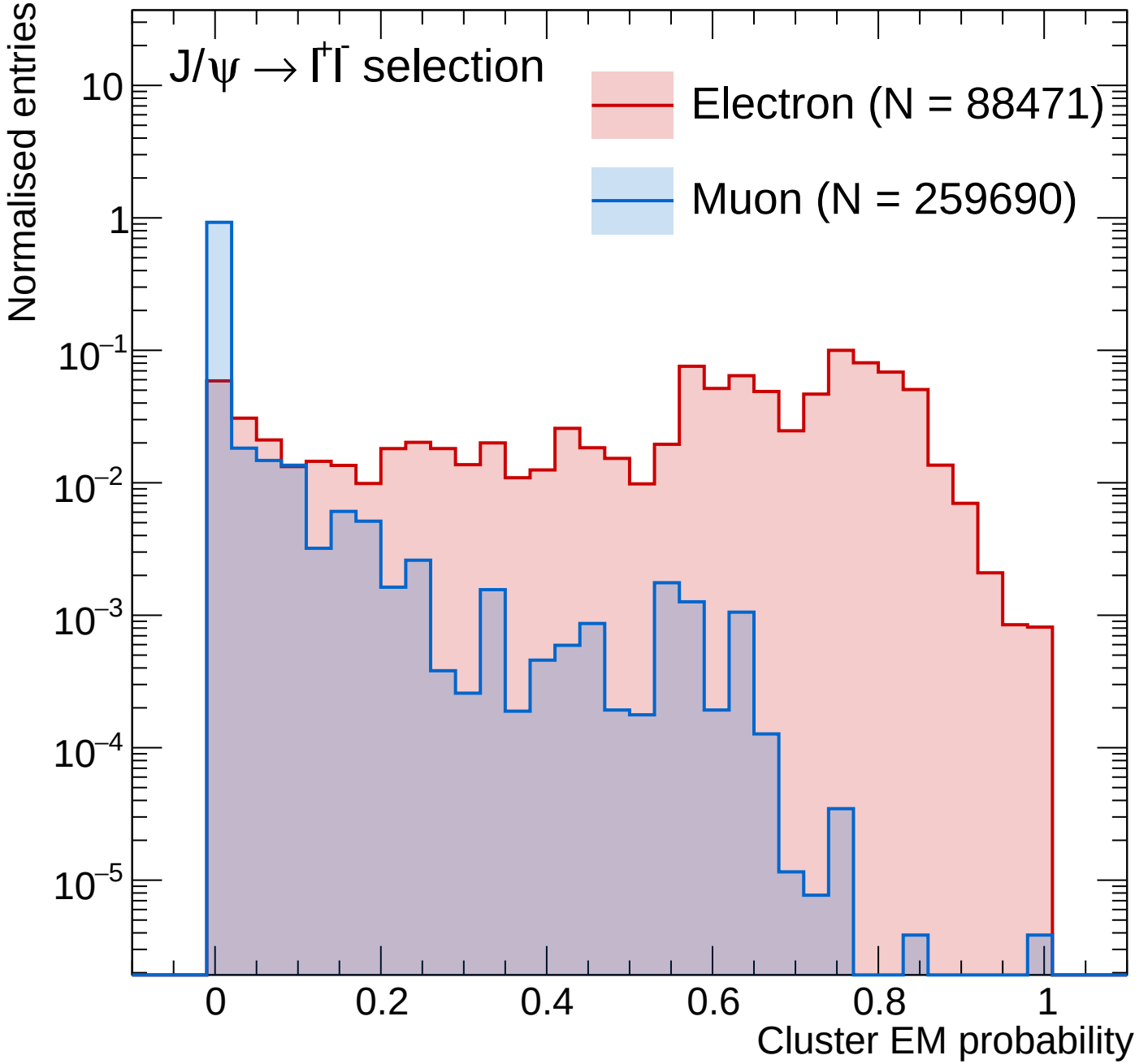

$J/\psi \rightarrow l^+l^-$ selection
Electron (N = 106122)
Muon (N = 287960)
Normalised entries
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Pixel dE/dx

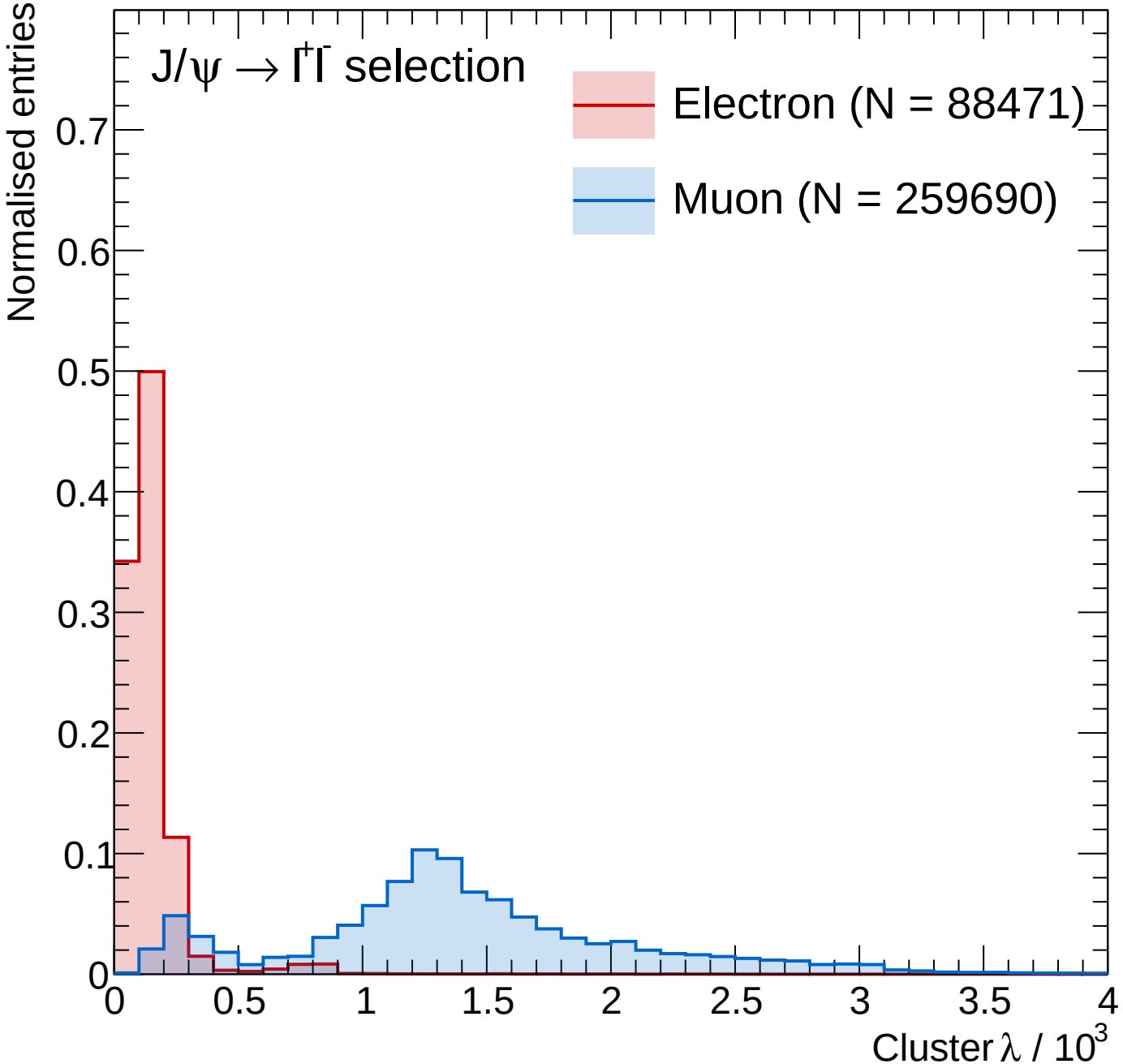

J/ψ → l+l- selection
Electron (N = 53061)
Muon (N = 143980)
Normalised entries
0
0.02
0.04
0.06
0.08
0.1
0.12
2.5
2.6
2.7
2.8
2.9
3
3.1
3.2
3.3
3.4
3.5
m_ll [GeV]

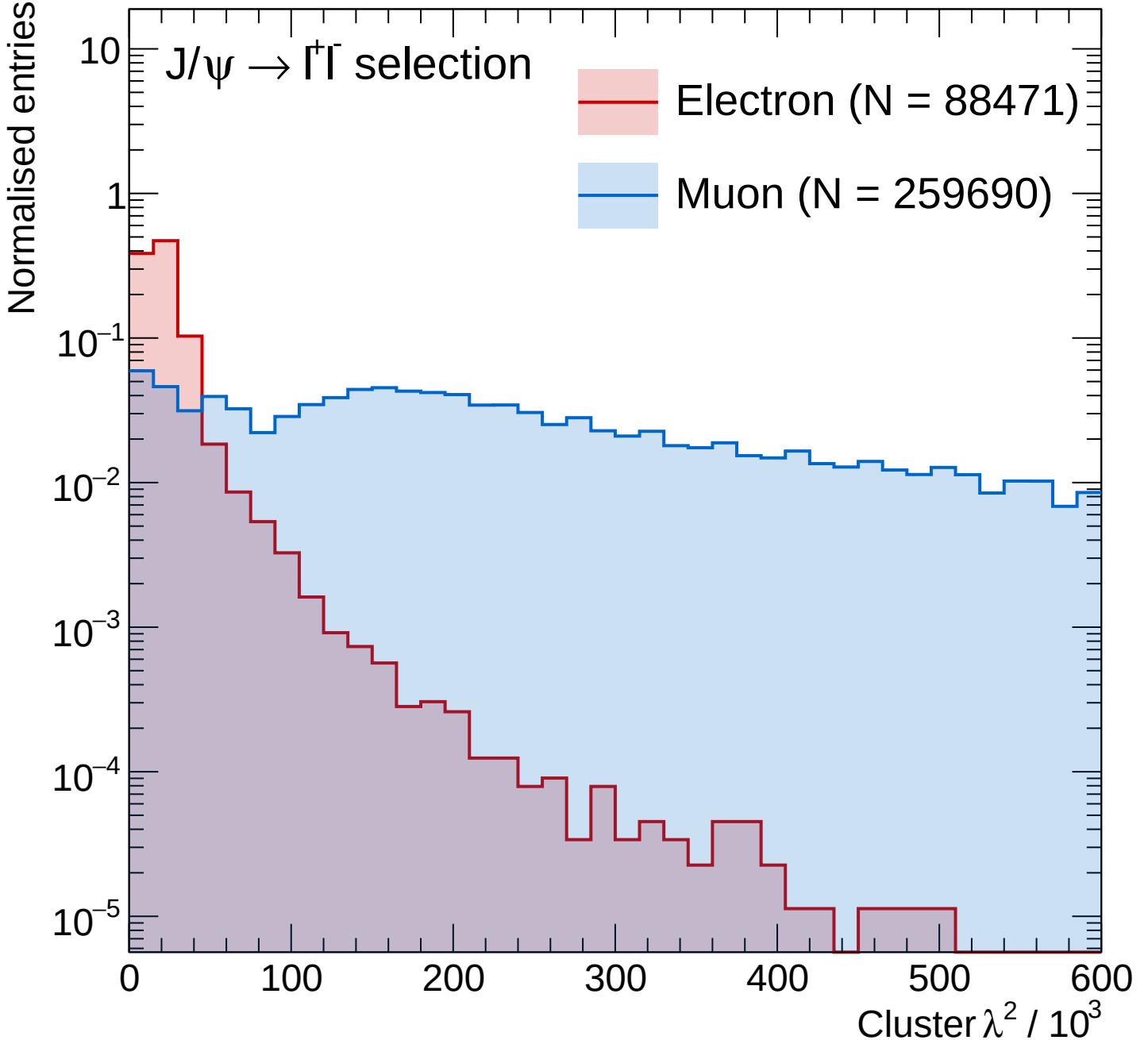

J/ψ → l⁺l⁻ selection
Electron (N = 106122)
Muon (N = 287960)
Normalised entries
10
1
10⁻¹
10⁻²
10⁻³
10⁻⁴
0
1
2
3
4
5
6
Number of matched topoclusters

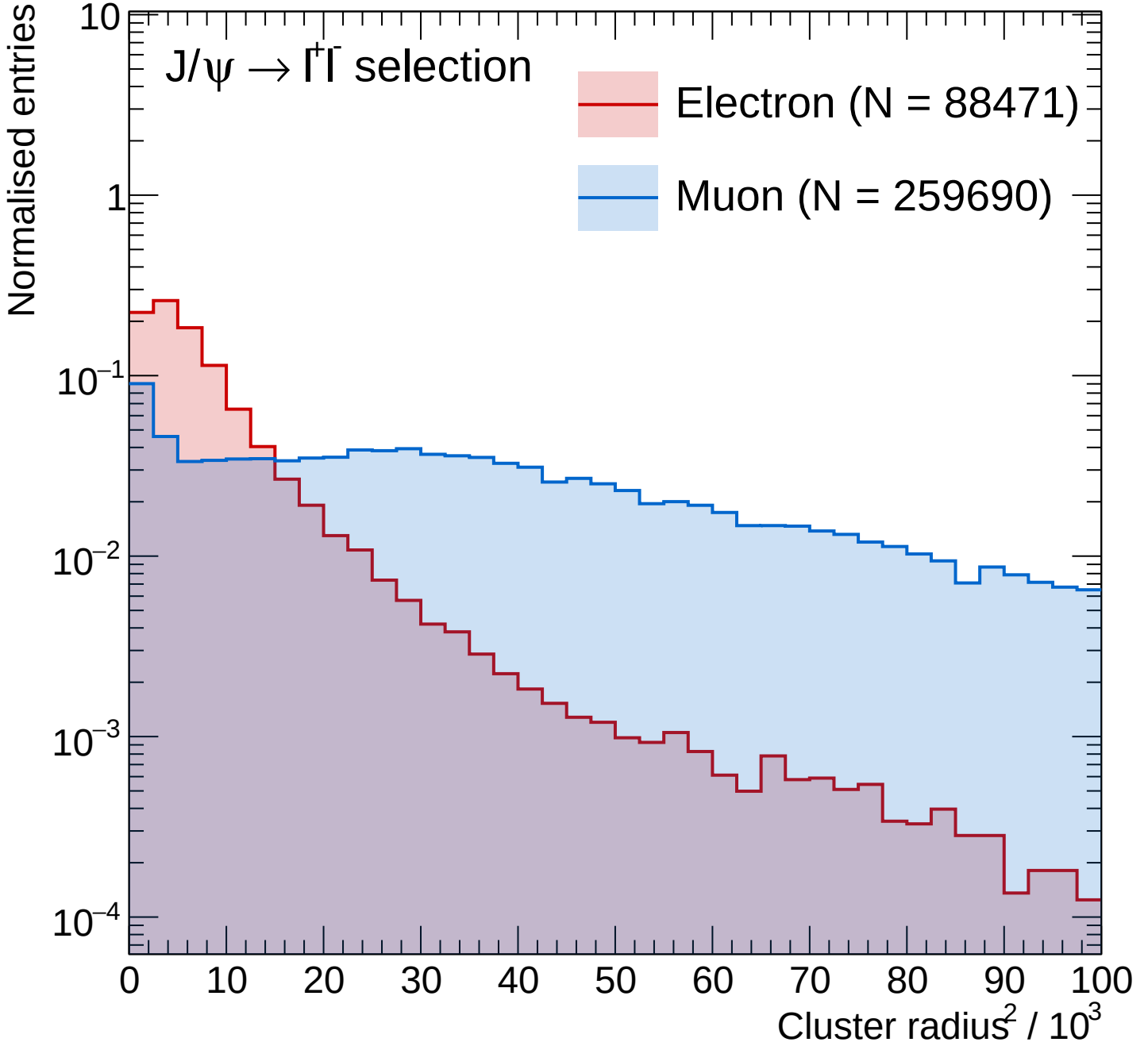

J/ψ → l⁺l⁻ selection
Electron (N = 88471)
Muon (N = 259690)
Normalised entries
0.045
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
0
0.5
1
1.5
2
2.5
3
3.5
Cluster p_T [GeV]

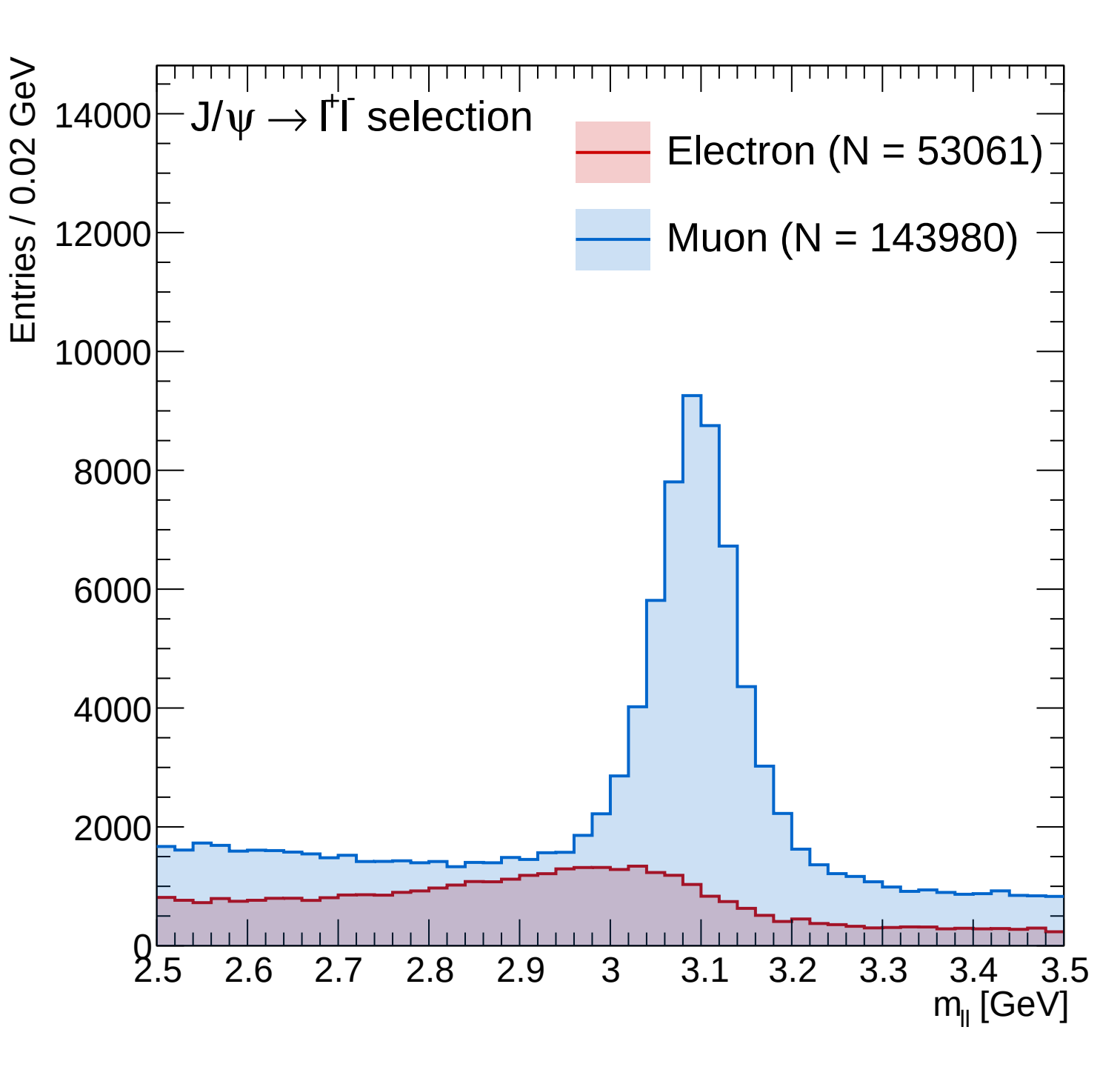

J/ψ → l+l- selection
Electron (N = 88471)
Muon (N = 259690)
Normalised entries
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
0
0.5
1
1.5
2
2.5
3
3.5
4
F = p_T^cl / p_T^trk

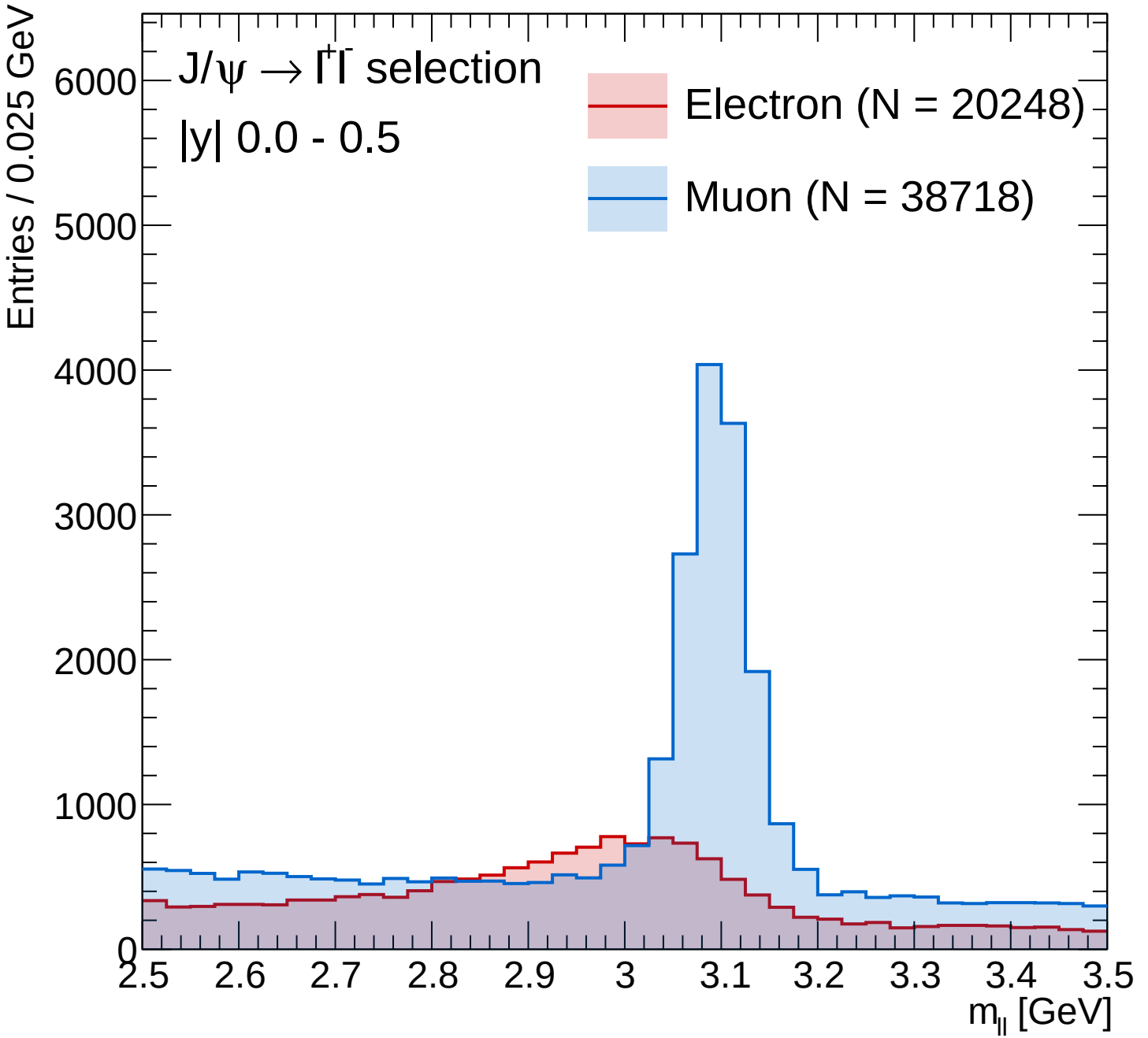

J/ψ → l⁺l⁻ selection
Electron (N = 88471)
Muon (N = 259690)
Normalised entries
10
1
10⁻¹
10⁻²
10⁻³
10⁻⁴
10⁻⁵
0
0.2
0.4
0.6
0.8
1
Cluster EM probability

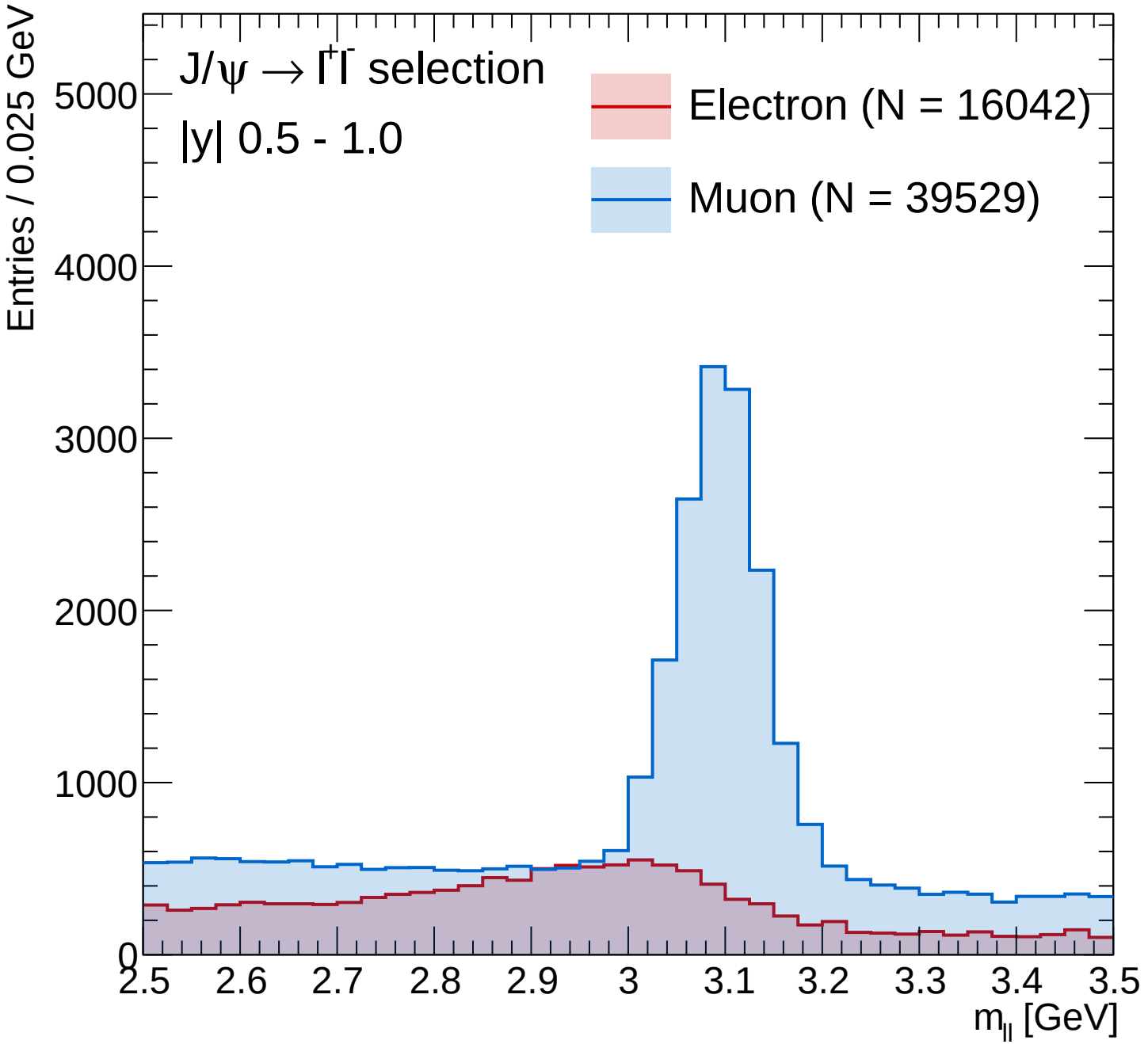

J/ψ → l+l- selection
Electron (N = 88471)
Muon (N = 259690)
Normalised entries
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0
0.5
1
1.5
2
2.5
3
3.5
4
Cluster λ / 10^3

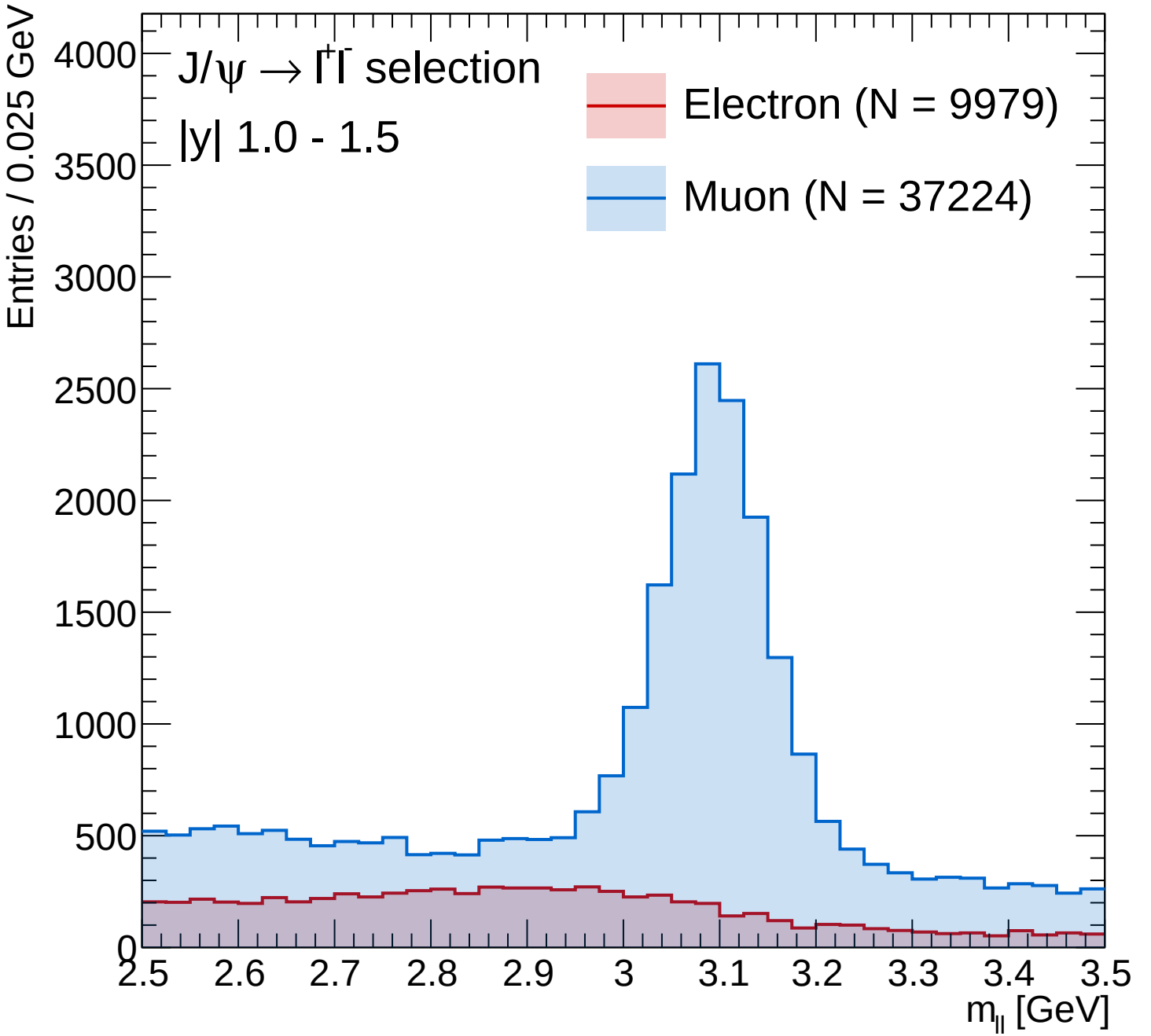

J/ψ → l⁺l⁻ selection
Electron (N = 88471)
Muon (N = 259690)
Normalised entries
10
1
10⁻¹
10⁻²
10⁻³
10⁻⁴
10⁻⁵
0
100
200
300
400
500
600
Cluster λ² / 10³

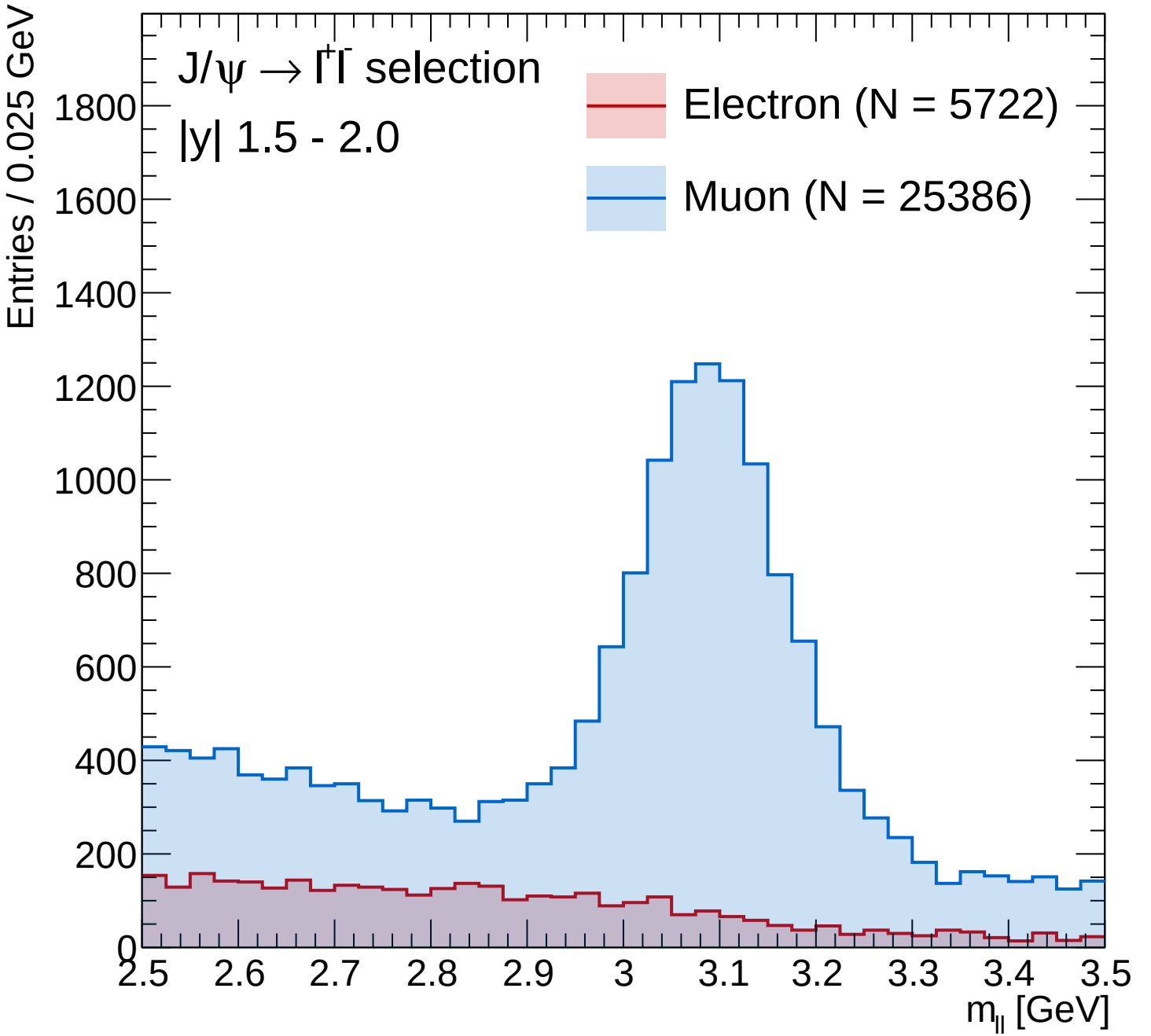

J/ψ → l+l- selection
Electron (N = 88471)
Muon (N = 259690)
Normalised entries
10
1
10−1
10−2
10−3
10−4
0
10
20
30
40
50
60
70
80
90
100
Cluster radius2 / 103

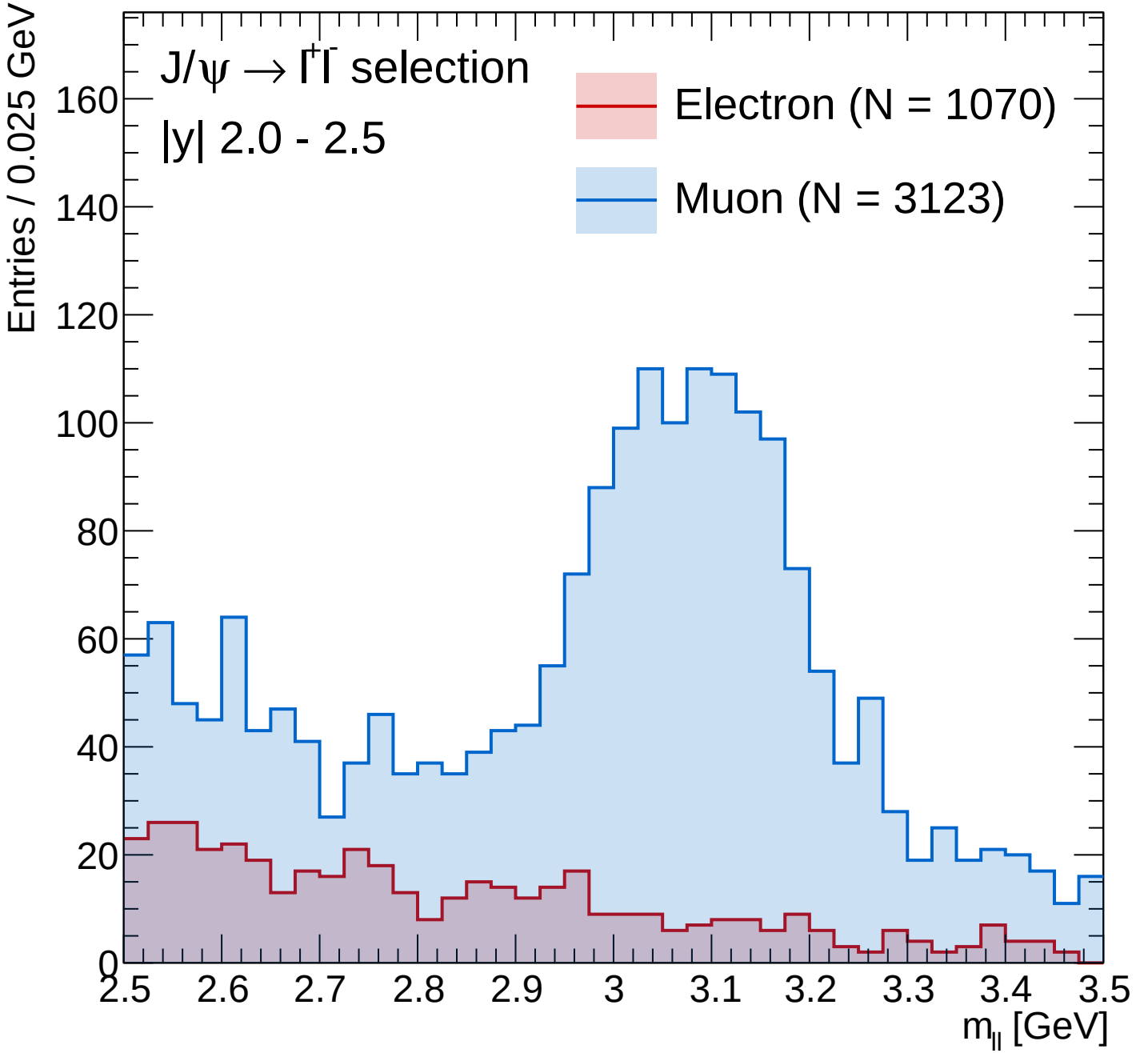

J/ψ → l⁺l⁻ selection
Electron (N = 53061)
Muon (N = 143980)
Entries / 0.02 GeV
14000
12000
10000
8000
6000
4000
2000
0
2.5
2.6
2.7
2.8
2.9
3
3.1
3.2
3.3
3.4
3.5
m_ll [GeV]


J/ψ → l⁺l⁻ selection
|y| 0.0 - 0.5
Electron (N = 20248)
Muon (N = 38718)
Entries / 0.025 GeV
6000
5000
4000
3000
2000
1000
0
2.5
2.6
2.7
2.8
2.9
3
3.1
3.2
3.3
3.4
3.5
m_ll [GeV]


J/ψ → l+l- selection
|y| 0.5 - 1.0
Electron (N = 16042)
Muon (N = 39529)
Entries / 0.025 GeV
5000
4000
3000
2000
1000
0
2.5
2.6
2.7
2.8
2.9
3
3.1
3.2
3.3
3.4
3.5
m_ll [GeV]


J/ψ → l⁺l⁻ selection
|y| 1.0 - 1.5
Electron (N = 9979)
Muon (N = 37224)
Entries / 0.025 GeV
4000
3500
3000
2500
2000
1500
1000
500
0
2.5
2.6
2.7
2.8
2.9
3
3.1
3.2
3.3
3.4
3.5
m_ll [GeV]


J/ψ → l⁺l⁻ selection
|y| 1.5 - 2.0
Electron (N = 5722)
Muon (N = 25386)
Entries / 0.025 GeV
0
200
400
600
800
1000
1200
1400
1600
1800
2.5
2.6
2.7
2.8
2.9
3
3.1
3.2
3.3
3.4
3.5
m_ll [GeV]


$J/\psi \to l^+l^-$ selection
|y| 2.0 - 2.5
Electron (N = 1070)
Muon (N = 3123)
Entries / 0.025 GeV
0
20
40
60
80
100
120
140
160
2.5
2.6
2.7
2.8
2.9
3
3.1
3.2
3.3
3.4
3.5
m_{ll} [GeV]